UNIT-2 I·P·IIND

ONESHOT CRASH COURSE

B PHARM 5TH SEM

IMPORTANT QUESTIONS

(1). Discuss in details about formulation of tablet (Intro) (prepn, production, manufacturing + machine used).
- Granulation method (wet + dry)
- Processing problems or manufacturing defect (resolve them).

(2). Define tablet coating, write the types of coating. Note on:-
- Step involved in sugar coating with eg. (different stages).
- film coating composition (various components) + Coating Pan.

(3). Describe the various parameter of the in-process & finished products quality control tests for tablets (coated tablets).

(4). Define and classify Liquid Orals. It's ideal characteristics. Explain formulation, evaluation, and manufacturing consideration of syrups, elixirs, suspensions and emulsions.
(it's evaluation) → challenge faced in development of oral suspension.

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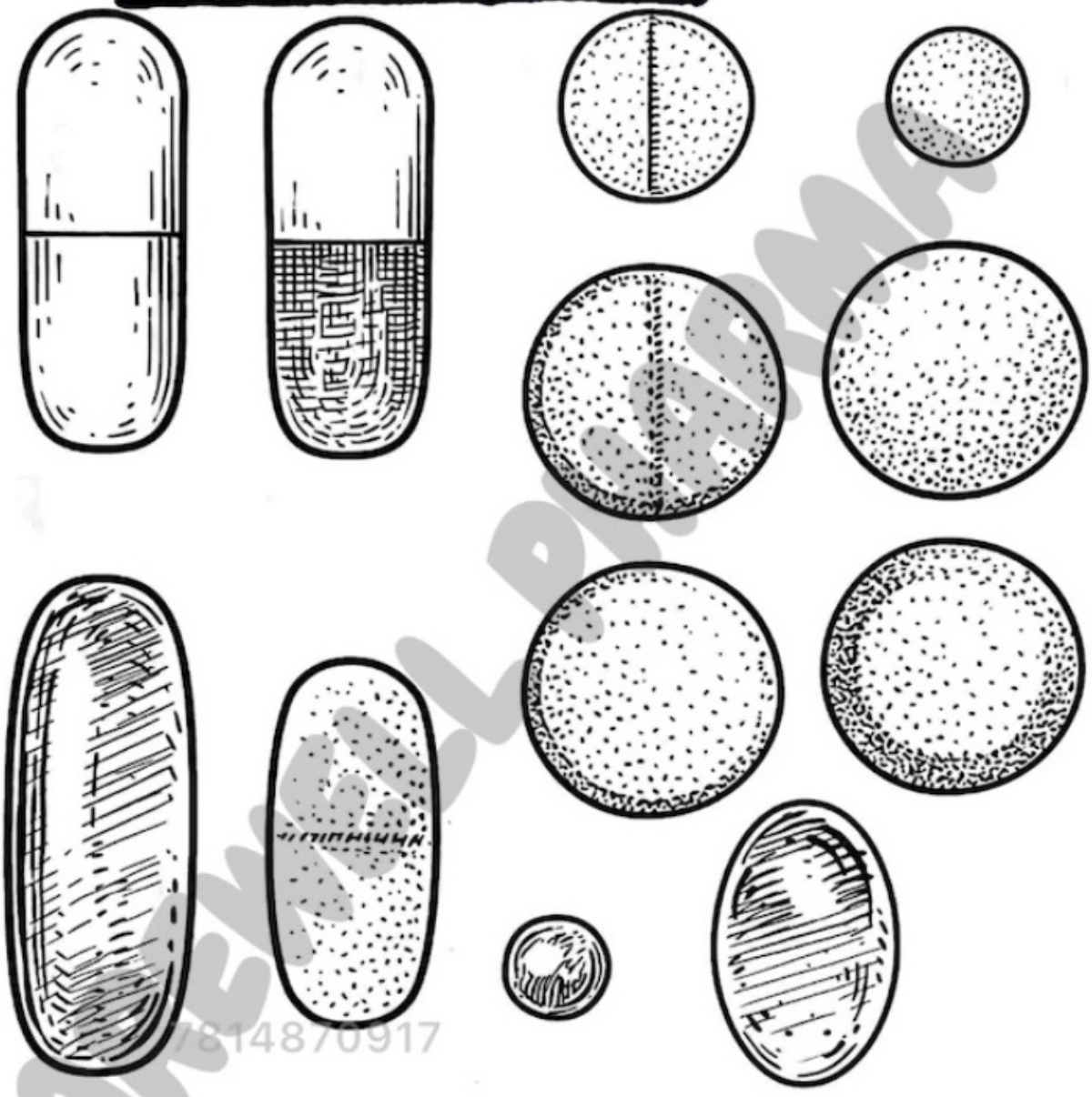
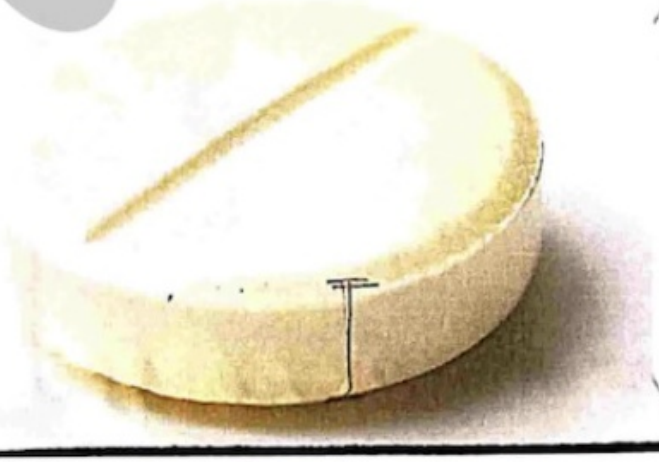
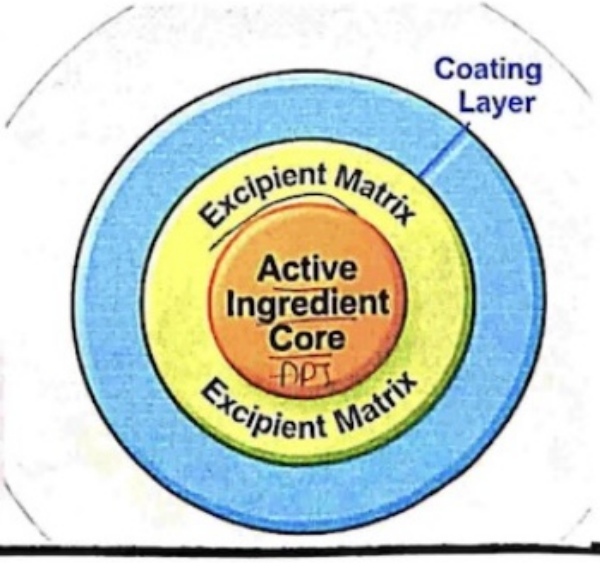
TABLETS







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① Discuss in details about formulation of tablet

- Granulation Method
- processing problems or manufacturing defects.
- Excipients used in tablet formulation.

Tablet

A tablet is a solid unit dosage form containing one or more active pharmaceutical ingredients (API) with or without excipients.

- It is usually Circular, oval or oblong in shape and prepared by compression or sometimes by molding.

Tablet → Solid unit DF.

API/medicant → Active Pharmaceutical Ingredients → Effect on body

+ Excipients → helps in the formulation of medicines (dosage form)

Tablets all shape image

- Tablets are the most common dosage form used worldwide because they are easy to manufacture, stable and convenient for patients.

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Advantage of Tablets

- Easy to handle, store and carry.
- Each tablet contains a precise quantity of drug.
- Simple to swallow, chewable or dispersible.
- Taste masking possible.
- More stable than liquid dosage forms.
- Easy and economical using machines.
- Attractive shapes, colors and flavors improve acceptability.

Disadvantages

- Difficult to swallow for children, elderly and unconscious patients.
- Can't easily reduce or increase amount.
- Not used in emergency (slow action).

Ideal Properties of Tablets

- Have uniform weight, size and appearance.
- Contain the correct dose of API.
- Have good stability.
- Show rapid disintegration and dissolution.
- Be economical to produce
- Be safe and effective for the patient.
- Be mechanically strong
- Have good taste, no irritation, acceptable color/appearance.

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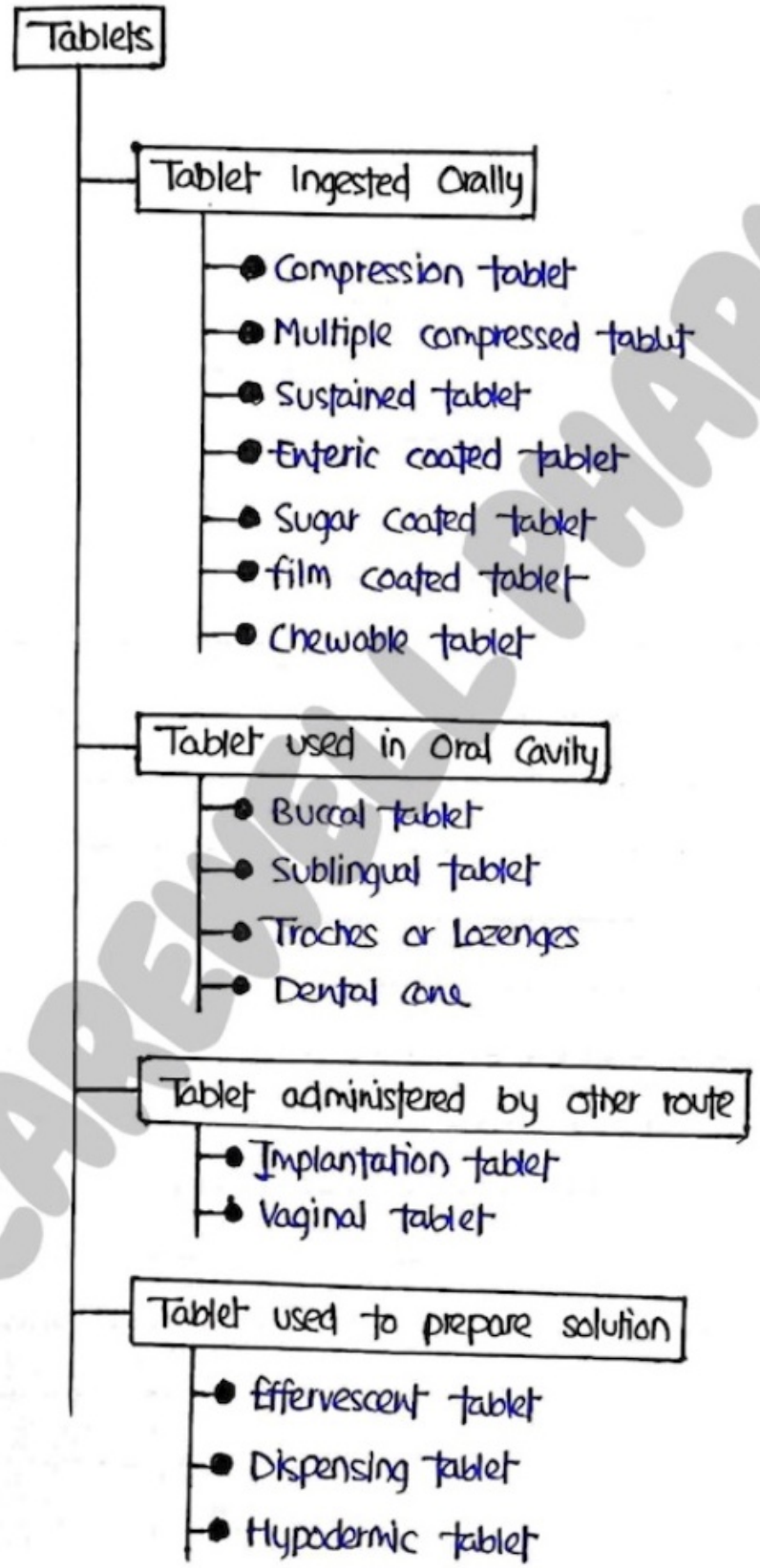
TYPES/CLASSIFICATION OF TABLETS



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1 Tablets Administered by Oral Route

These are those tablets which are introduced in our body through oral route or swallowed with water.

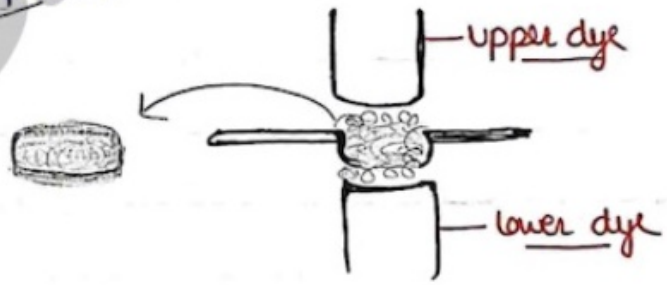
(i) **Compressed Tablets** - These are those tablets which are generally prepared by compression of powdered, granulation material with the help of punching machine.

- Simple tablets without any coating, directly compressed.
- Provide systemic action after swallowing.

Method of Preparation - Wet granulation, dry granulation or direct compression.

Example - Paracetamol, Metformin.

Tablets



(ii) **Multiple compressed tablet** - These are those tablets which are composed of two or more different layer.

- These are used for those tablet which has two or more incompatible substance with each other.
- Used to separate incompatible drugs, controlled release.
- These tablets can be prepared by more than a single compression.

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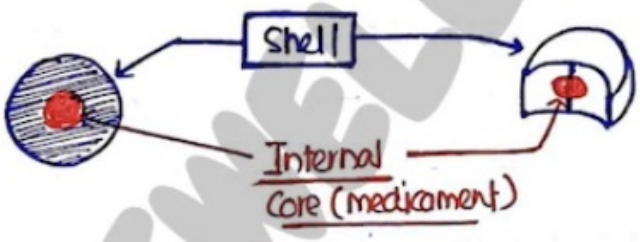
Method of Preparation - Compression in multiple stages using rotary press.

Example - Antihypertensive combinations.

(a) Multilayer or layered tablet - They contain either two or more layered (components).

(b) Compressed coated tablet - They contain two parts internal core (small tablet) and surrounding coat which act as shell.



(iii) Sustained tablet - These tablet are prepared for the purpose of long duration of action.

- So these tablets release their Active ingredient over prolonged period.

Method of Preparation - Hard compression.

Example - Metformin SR (used in diabetes), Diclofenac SR (used in pain/inflammation)

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(iv) **Enteric coated tablet (ECT)** - These are those tablets which are resist disintegration in stomach, dissolve in intestine.

- These tablet are prepared to protect drug from stomach acid or protect stomach from irritating drugs.
- It has delayed release properties.

Method of Preparation - Coating with pH - sensitive polymers [Eudragit, Cellulose acetate phthalate).

Example - Omeprazole, Aspirin EC.

(v) **Sugar Coated tablet** - These are those tablet which is covered with sugar layers to improve taste and appearance.

- It is used to mask bitter taste, protect drug, increase patient compliance.
- Also protect drug and enchance appearance.

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Method of Preparation - Core tablet → Sugar syrup coating → drying (repeated cycles).

Example - Multivitamin sugar coated tablets.

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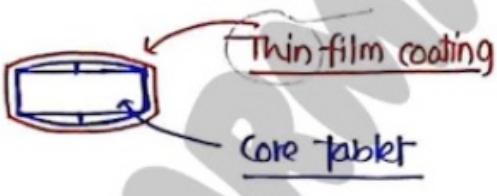
(vi) **Film Coated tablet** - These are compressed tablet coated with a thin layer of polymer film, forming a skin like film over the tablet.

- It is used to protect from light, moisture and mask unpleasant taste.

Eg. Antibiotics etc..



Polymers ⇒ Hydroxy Propyl Cellulose etc..

Method of Preparation - Core tablet sprayed with polymer solution (Eg. HPMC, PVP) → drying.

Example - Diclofenac sodium (Voveran).

(vii) **Chewable tablet** - These are those which are designed to be chewed before swallowing.

- It is useful for children, elderly, antacids, vitamins.

→ Digene / Antacid..

Method of Preparation - Direct compression with mannitol / sorbitol for chewable texture.

Example - Digene, Vitamin C chewable.

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2 Tablets Administered in Oral Cavity.

A tablet used in the oral cavity is typically designed to dissolve or disintegrate in the mouth rather than being swallowed whole.

(i) **Buccal Tablets** - These are those tablets which are placed in cheek pouch, drug absorbed through buccal mucosa. (between gums and lips) where they dissolved or disintegrate slowly and absorbed into blood without passing GIT.

- It is used to provide sustained drug release.

Method of Preparation - Direct compression with mucoadhesive polymers (Carbopol, HPMC).

Example - Buprenorphine buccal tablets.

(ii) **Sublingual tablets** - These are those tablet which are placed under tongue for rapid absorption into blood without passing GIT.

- It is used for Emergency drugs, bypass liver metabolism.

Method of Preparation - Compression with soluble excipients.

Example - Nitroglycerin (Vasodilator in Angina (CAD)), Isosorbide dinitrate.

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(iii) **Lozenges / Troches** - These are those tablets which are commonly used to treat sore throat or to control coughing in common cold.

- These tablets are intended to dissolve slowly in mouth for local action.

Method of Preparation - Molding or compression of sugar - based mass.

Example - Strepsils, Clotrimazole lozenges.

(iv) **Dental Cone** - These are those tablet which are small sterile tablets placed in tooth socket. after extraction, prevent from bacteria.

Method of Preparation - Sterile molding or compression under aseptic conditions.

Example - Tetracycline dental cones.

dental cone

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3. Tablets Administered by Other Routes.

These are those tablets which are administered other than oral routes. + Rectal tablets ~

(i) **Implantation tablets** - These are those tablets which are sterile and inserted beneath the skin, release drug for long periods.

- These tablets have slow absorption and provide continous release of drug for the months or a year.
- These tablets provide prolonged drug release.

Method of Preparation - Compressed under sterile conditions, usually cylindrical in shape.

Example - Testosterone implants, Norethisterone implants.

(ii) **Vaginal Tablets** - These are those tablets which are intended for local effect in vagina. and dissolve slowly.

- These tablets are used to release antibacterial agents, antiseptic and contraceptive purposes.

Method of Preparation - Compressed with excipients that dissolve in vaginal fluid.

Example - Clotrimazole vaginal tablets.~

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4. **Tablet used to prepare solution** -

These are those tablets which firstly used to prepare solution than these solution used for action.

(i) **Effervescent tablet** - These are those tablet which dissolved in water and release CO_2 gas.

- Release of Co_2 in water produce disintegration of tablet & produce (increase) palatability and also give rapid action

Method of preparation - Compression of granular effervescent salts.

Example - ENO, Disprin etc--

(ii) **Dispensing tablet** - These are those tablets which are completely dissolve in water before administration.

- This is because tablet contain large amount of highly potent APIs required water to make solution.
- It is useful for children, patients needing solution dosage.

Method of Preparation - Direct compression with soluble excipients.

Example - Vitamin C soluble tablet.

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(viii) **Hypodermic Tablet (Obsolete)** - These are those tablets which are earlier used to prepare injectable solution.

- It is useful in Emergency injectable preparation.

Method of Preparation - Sterile compression under aseptic conditions.

Example - Morphine hypodermic tablets. (Historical use).

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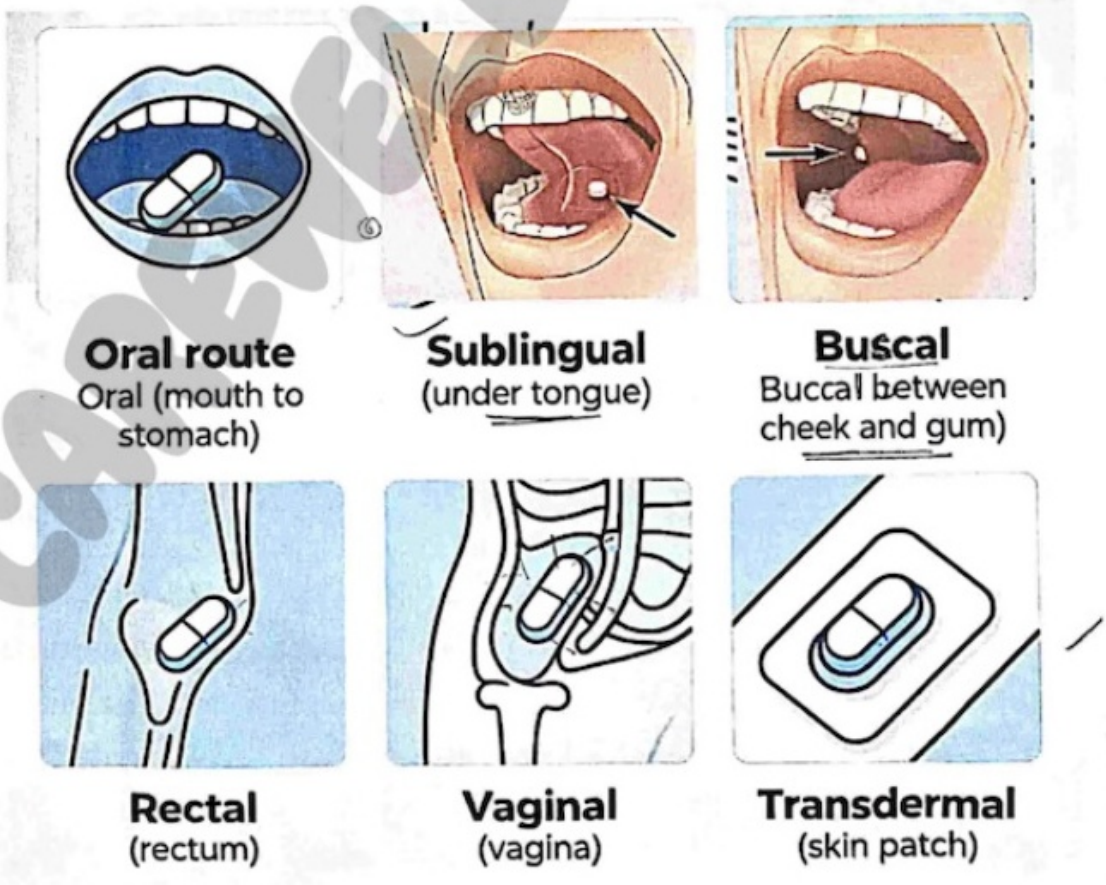

EXCIPIENTS USED IN THE FORMULATION OF TABLETS

Excipients - An excipients are pharmaceutically inactive substance added along with the Active Pharmaceutical Ingredients (API) during tablet formulation.

- These excipients has (no) therapeutic value & inert material but are added with drug in formulation for –
  - improve stability & bioavailability.
  - improve patients acceptability
  - help in (production)
  - improve flow of powder.

  } formulation

Ideal Properties of Excipients

- Inert (no pharmacological activity)
- Non-toxic & non-irritant.
- Chemically (stable) (not react with drug)
- Economical & easily available
- Should not affect bioavailability
- Compatible with (API) & other excipients.
- Good organoleptic properties
- Allow large-scale manufacturing (good flow, compressibility).

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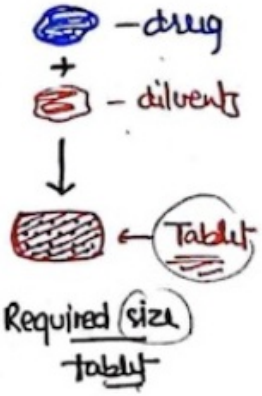
- Following excipients are used in tablet formulation :-
  (i) Diluents / Filler (Bulking agents)
  (ii) Binders / Adhesives
  (iii) Disintegrants
  (iv) Glidants
  (v) Lubricants
  (vi) Anti-adherents
  (vii) Coloring Agents
  (viii) Flavoring Agents
  (ix) Sweeteners
  (x) Preservatives
  (xi) Adsorbing agents.

(i) **Diluents / fillers** - Also known as **Filler** or **Bulking agents**. These are designed to increase bulk of tablet, when drug dose is very small.

Ideal Properties - Inert, stable, good compressibility, non-hygroscopic.

Uses - Provide uniform size and weight
- Improve flow and compressibility.

Example - Lactose, Starch, Mannitol etc...



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(ii) **Binders** - These are those substance which helps to bind powder particles together to form granules, improving mechanical strength of tablet.

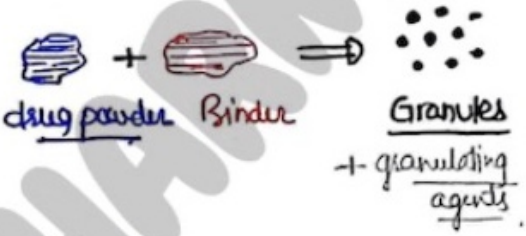
Binding agents

- These agents are present in both (dry) or in (wet) form.

dry granulation — wet granulation

Ideal Properties -

- Non-toxic, stable
- should not affect drug release.



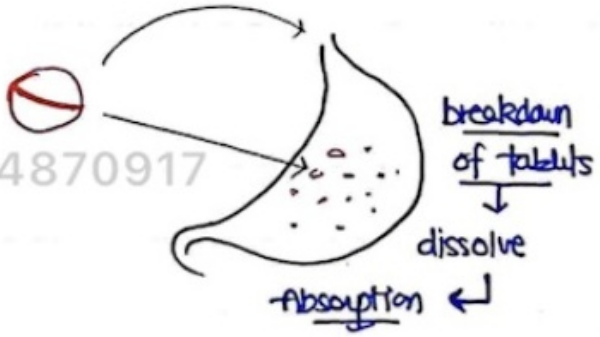
Uses - Provide cohesion, prevent tablet breabage.

Example - Acacia, Gelatin, Starch paste etc...
Tragacanth,

(iii) **Disintegrating agents** - These are those substance which are (Facilitate) breakup of tablet into smaller particles after swallowing → enhances dissolution and absorption.

Ideal Properties -

- Swell rapidly in water
- Non toxic
- Effective in small amounts.



Uses - Ensure rapid disintegration and drug release.

Example - Sodium starch glycolate, Bentonite etc...

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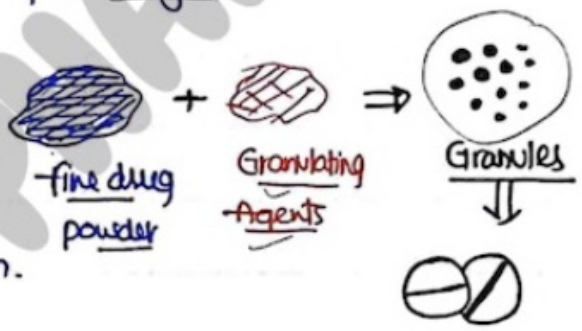
(iv) **Granuling Agents** - These are those substance that hold the powder particles together to form granules.

- These agents provide cohesiveness, improve flow properties and give mechanical strength to tablets.
- These agents provide proper (moisture) to convert fine powder into (damp) mass which after passing through a (sieve) of suitable number forms granules.



Ideal Properties -

- sufficient binding strength.
- effective at low concentration.

Uses • Providing strong binding and stable formulation.

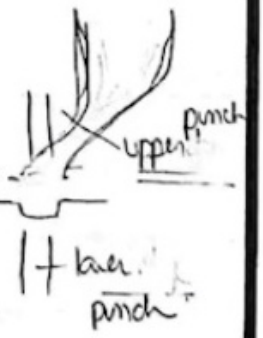
- Improve (taste), appearance.

Example - Diclofenac, Paracetamol.

(v) **Glidants** - These are those agents which improve (flow) of powder material.

Ideal Properties - Effective at (low) concentration, chemically inert.

Uses - Prevent sticking of powder to hopper & dies.



Example - (Talc), Colloidal silicon dioxide.

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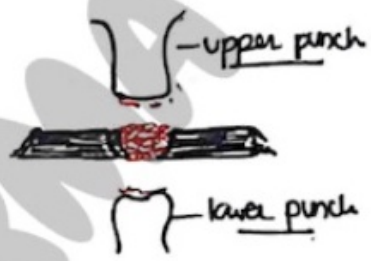
(v) **Lubricants** - These are those substance which prevent adhesion of tablet material to die walls & punches. Reduce friction during compression and ejection.



Ideal Properties -

- Effective at low concentration (0.5-2%)
- inert, stable.

Uses - Smooth ejection of tablets, prevent sticking.

Example - Magnesium stearate, Calcium stearate. Talc, stearic acid etc.

(vi) **Anti-adherents** - These are those substance which prevent sticking of tablet material to punch face.

Ideal Properties -

- inert, stable.

Uses • Improve production efficiency by reducing punch sticking.

Example • Talc, Starch, Magnesium stearate.

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(vii) **Colouring agents** - These are those substance which provide aesthetic appeal, aid in product identification. good/elegant appearance

Ideal Properties -

- Non - toxic, stable
- FDA / pharmacopoeia approved.

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Uses • Differentiate strengths, brand recognition.
- used to helps in identification of tablets.

Example - FD & C dyes, Titanium dioxide (white color).

(viii) **Flavoring Agents** - These are those substance which is used to improve taste, especially in chewable & ODTs.

Uses - Increase patient compliance. (children, elderly).

Example - Orange, vanilla, Peppermint. (vicks, ginger, menthol etc)

(ix) **Sweeteners** - These are those substances which improve palatability by masking bitter taste.
↑ acceptability

Uses - Mainly in chewables, ODTs, effervescent tablets.
orally disintegrating tablets

Example - Saccharin, Sorbitol, Mannitol, Sucrose.

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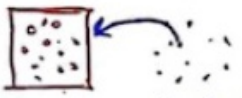
(X) **Preservatives** - These are those substance which prevent microbial growth in tablets containing moisture.

Uses - Enhance stability of formulations.
Example - Parabens, Methylparabens.

(xi) **Adsorbing Agents** - These are those substance which Absorb small amounts of moisture.

Uses - Protect moisture - sensitive drugs.
Example - Silica, Magnesium oxide.



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Formulation of Tablets.

Also known as (tablet) manufacturing process.

- It is the process of designing and preparing tablets by combining the Active Pharmaceutical Ingredients (API) with suitable (excipients).
- There are three methods for this :-

Granulation →
- (Wet) granulation
- (Dry) granulation
- (Direct) compression.

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Granulation - It is the process of (size) enlargement where fine powders of drug is converted into small (granules) which further used to prepare tablets through compression.

① **Wet granulation** - In this process, we use (binder) solution in (wet) form. The steps required for this :-

(i) Mixing of API & excipients.
(ii) preparing a (damp) mass
(iii) Screening - granulation
(iv) Drying
(v) Dry screening
(vi) Lubrication
(vii) Compression.

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(i) **Mixing** - Weigh the required amount of API (medicament) and all excipients.

- Now, these excipients such as diluents, and half amount of disintegrating agents are mixed with API through blender (V- cone blender).

(ii) **Preparing a damp mass** - In this step add the mixture to the solution of binder (in wet form) to make a damp mass.

(iii) **Wet screening** - The prepare damp mass is screened using seive with the help of hand or suitable equipment and forms granules.

(iv) **Drying** - These wet granules are dried in hot air oven at 60°C (adjust).

(v) **Dry screening** - After drying these granules are screened through a seive to get uniform size granules.

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(vi) **Lubrication** - Now these granules mixed with glidant then add lubricant in it and also add remaining disintegrants.

(vii) **Compression** - Now compress this granules into desired punching machine to give proper shape and size to the tablets.

Advantages -

- Produces good flow properties of granules.
- Uniform distribution of low - dose drugs.
- Have better appearance and mechanical strength.
- Can modify drug release by adjusting binder concentration.
- Reduces dustiness and segregation of powders.
- Need controlled particle size distribution.

Disadvantages

- Not suitable for moisture - sensitive drugs.
- Requires more equipment and manpower.
- Risk of Microbial contamination.

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② Dry granulation [Slugging / Double compression method]

In this process, we use binder in dry form.

The steps required for this :-

(i) Mixing
(ii) Slugging (first compression)
(iii) Crushing (breaking)
(iv) Screening
(v) Mixing
(vi) Compression.

(i) **Mixing** - Weigh the desired (required) amount of API and all excipients.

- Now add these excipients with API and half amount of lubricant with dry binder (in blender).

(ii) **Slugging** - Now these mixture are compressed into large flat tablets (pellets) known as "slug". (about 1 inch in diameter).

(iii) **Crushing** - These slugs again breaks into small particles using mill (suitable).

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(iv) **Screening** - These small particles screened to produce uniform sized granules.

(v) **Mixing** - Now add remaining amount of lubricant. And also add glidant and disintegrating agents to granules and mixed it into blenders.

(vi) **Compression** - Compress these granules into desired punching machine to give proper shape and size to the tablets and produce tablets.

Advantages -

- Suitable for moisture - sensitive drugs
- Faster and economical
- Have good stability
- Suitable for heat - sensitive drugs.
- Avoids problems of drying like over - drying or sticking.

Disadvantage -

- Poor mechanical strength.
- Expensive.

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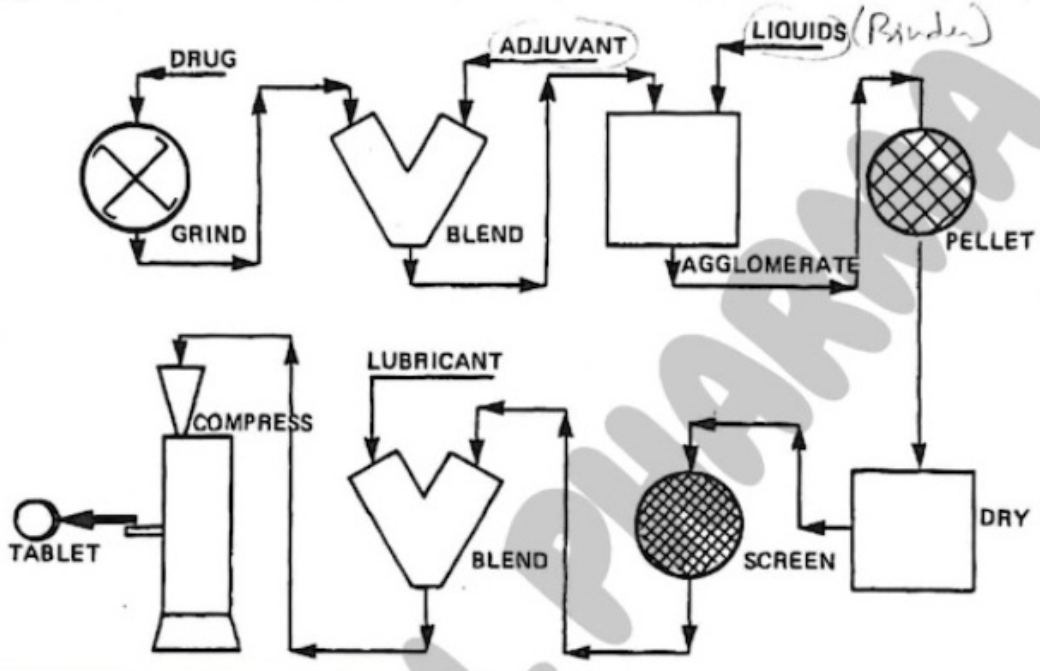
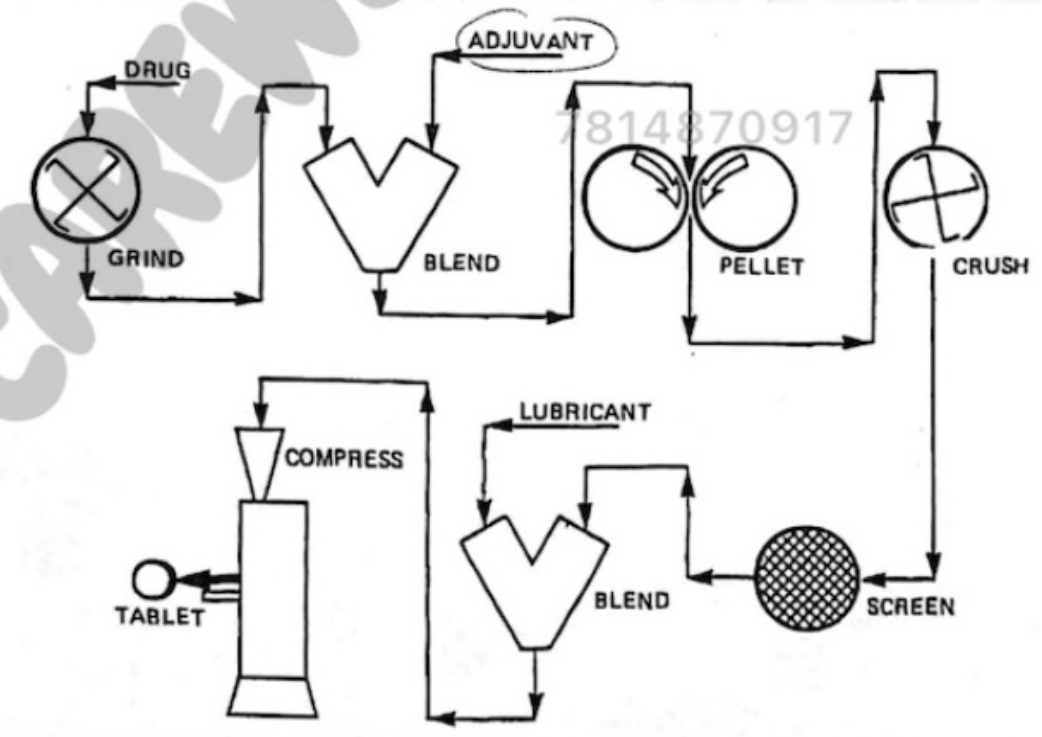
WET GRANULATION



DRY GRANULATION



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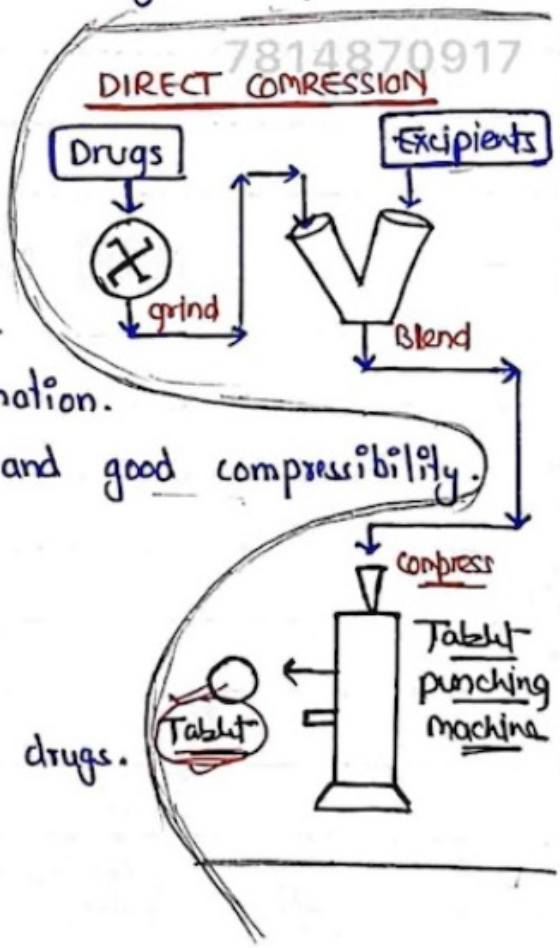
③ Direct Compression Method.

This is a tablet manufacturing technique in which the drug and suitable excipients are directly blended and compressed into tablets without any intermediate processing step like wet or dry granulation.

- It is considered the simplest, fastest and most economical method of tablet production.
- This method is possible only when the drug or excipients have excellent flowability and compressibility.

- Firstly weigh the API (medicament) and some suitable excipients such as Diluent, Disintegrats, lubricant etc...
- Then mixed and prepare tablets through compression.



Advantages

- Simple and time saving process.
- Suitable for heat-sensitive drugs.
- Less chance of error or contamination.
- Useful for drugs with high dose and good compressibility.
- Cost effective.

Disadvantages

- Not suitable for low-dose potent drugs.
- Poor tablet strength.

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TABLETS

Topic :→ Equipment used for manufacturing & Tablet processing Defects

Defects

Tablets — API + Excipients

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granules

convert granules to tablets

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Tablets

through compression

Tablet compression Machine

equipments

① Single punch tablet machine

↓

for small scale,

eg labs, experiment

single tablets at a time

② Multi punch Rotary tablets machine

↓

for large scale

eg Industries

multiple tablet at a time

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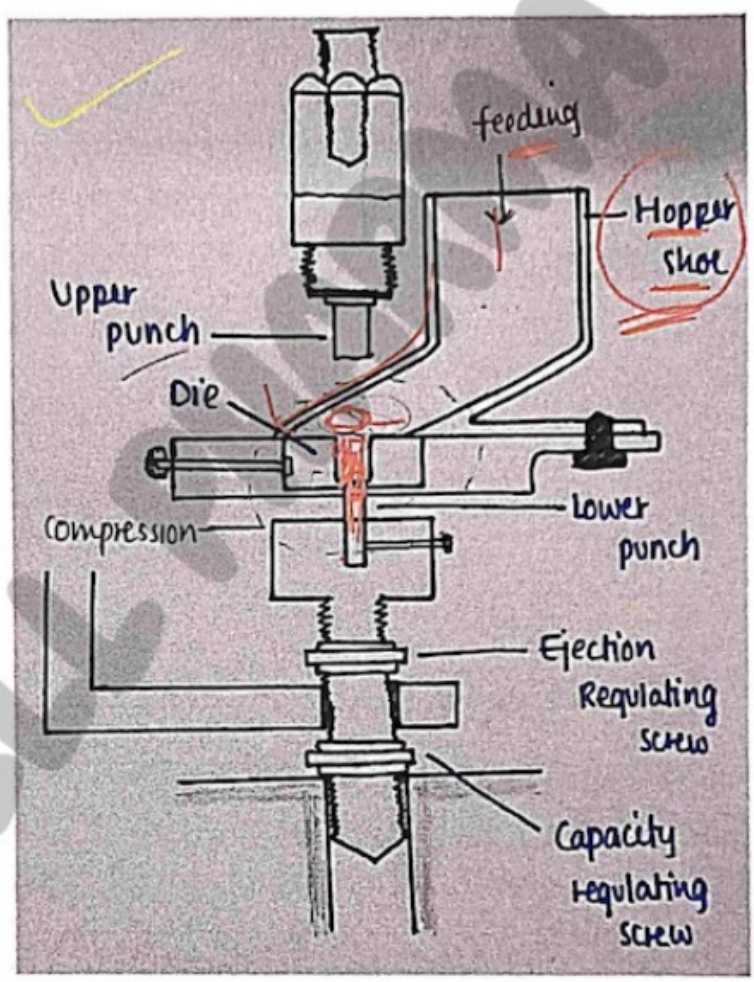
(1) (Single) punch tablet machine

- Hopper shoe
- Die
- Die Cavity
- Punch < upper / lower
- Cam tracks



WORKING

- feeding / filling of granules
- Compression
- Ejection of tablet

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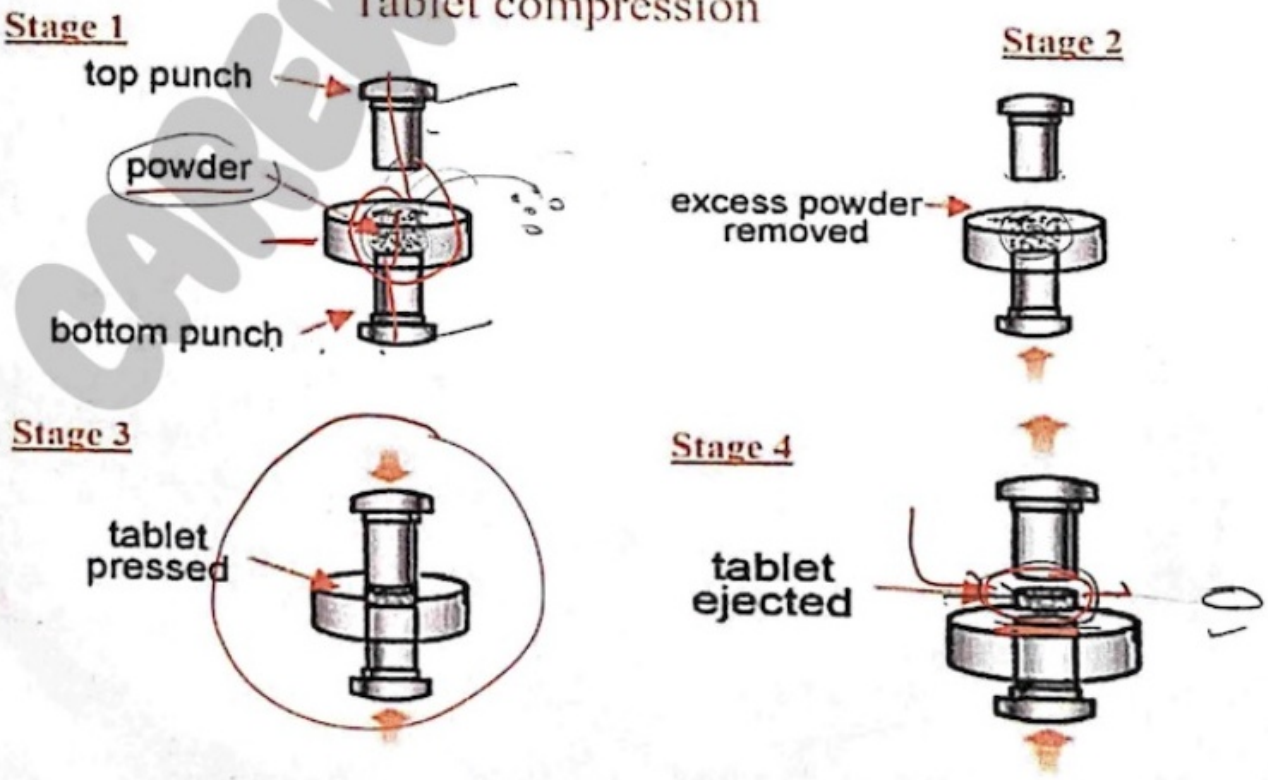
Single punch tablet machine

Tablet compression



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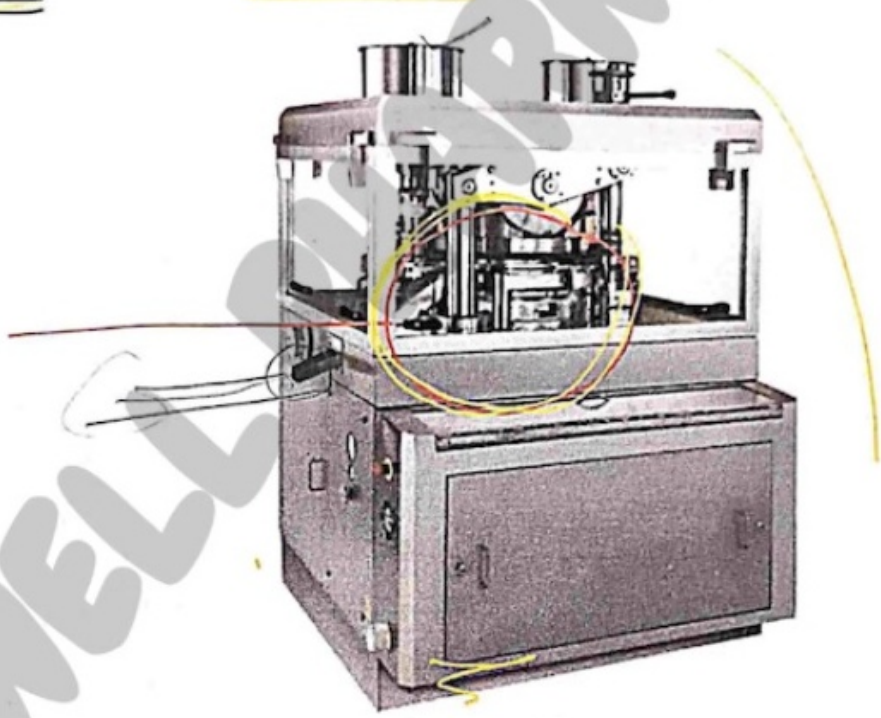
(2) Multipunch Rotary tablets machine

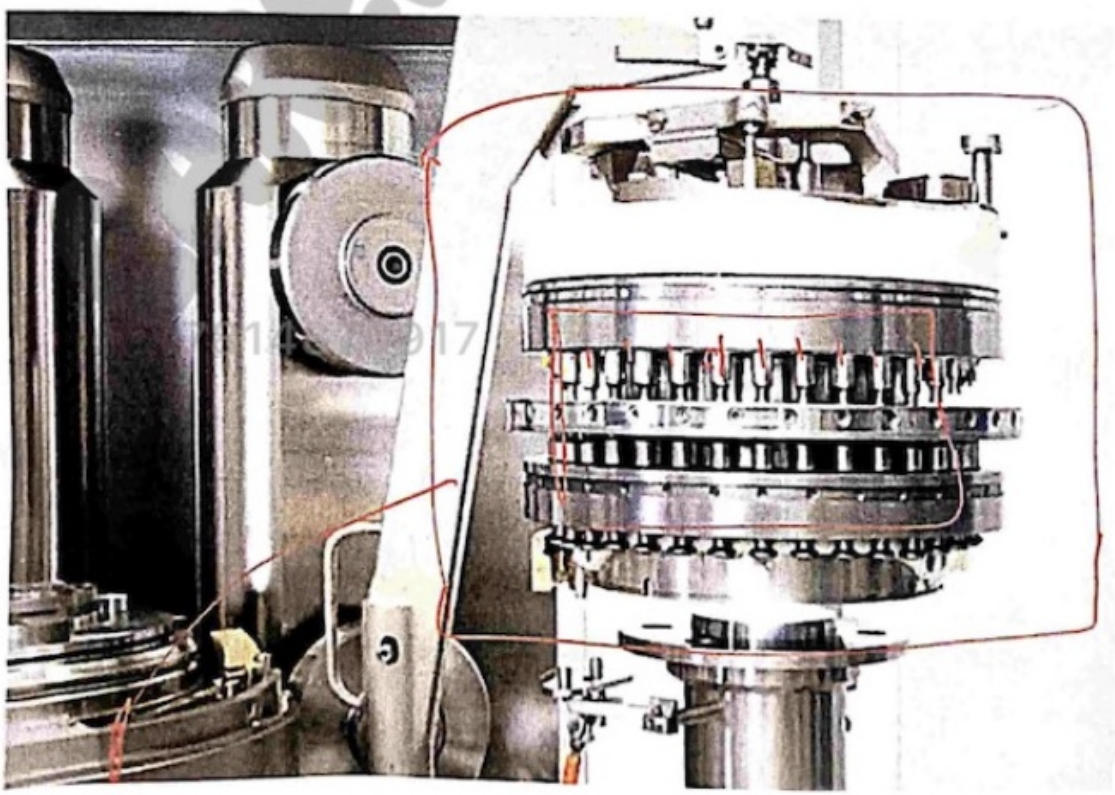
more than 1 set of punch

Rotation in machine

1200 / min

Zoom





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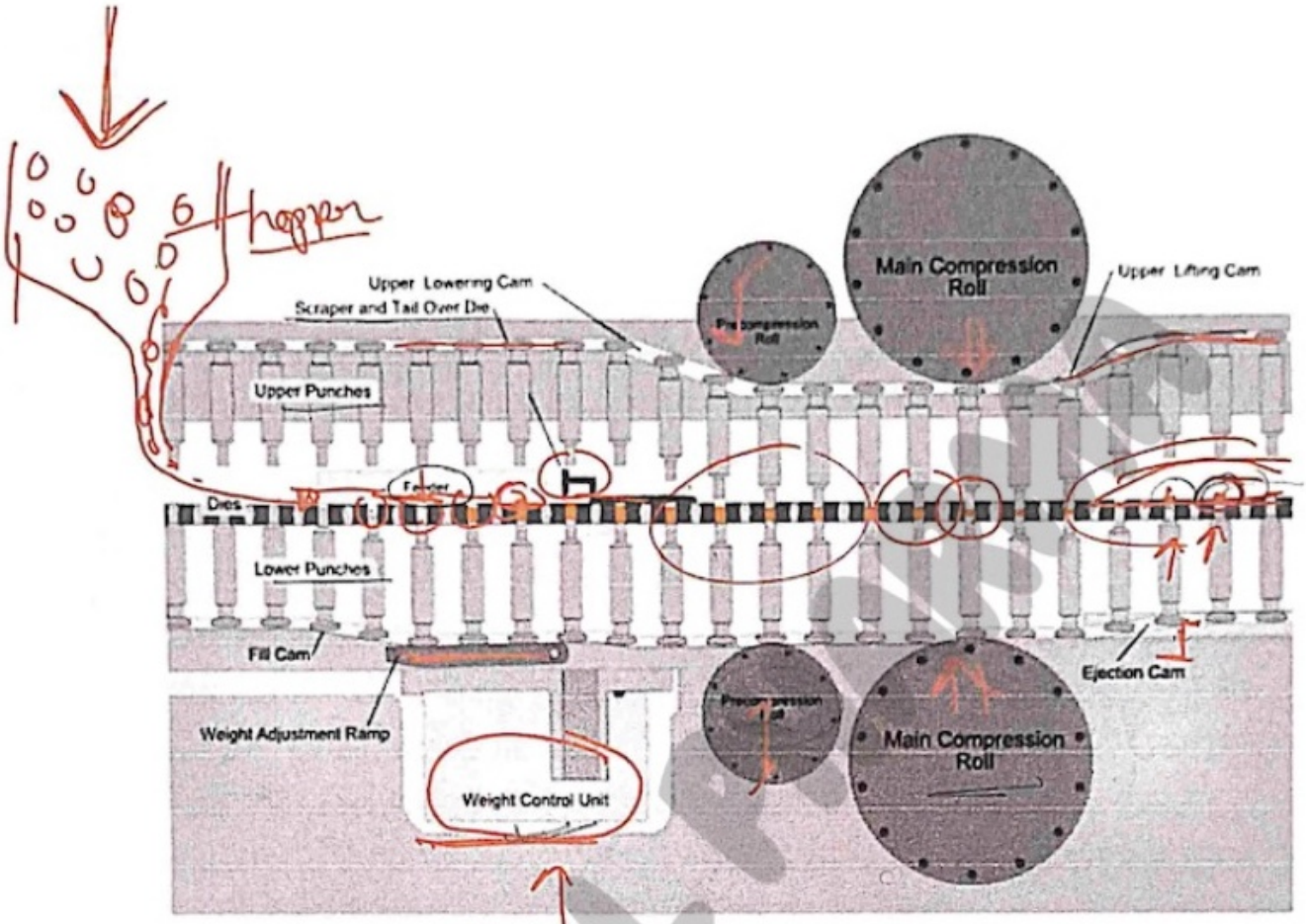

Construction

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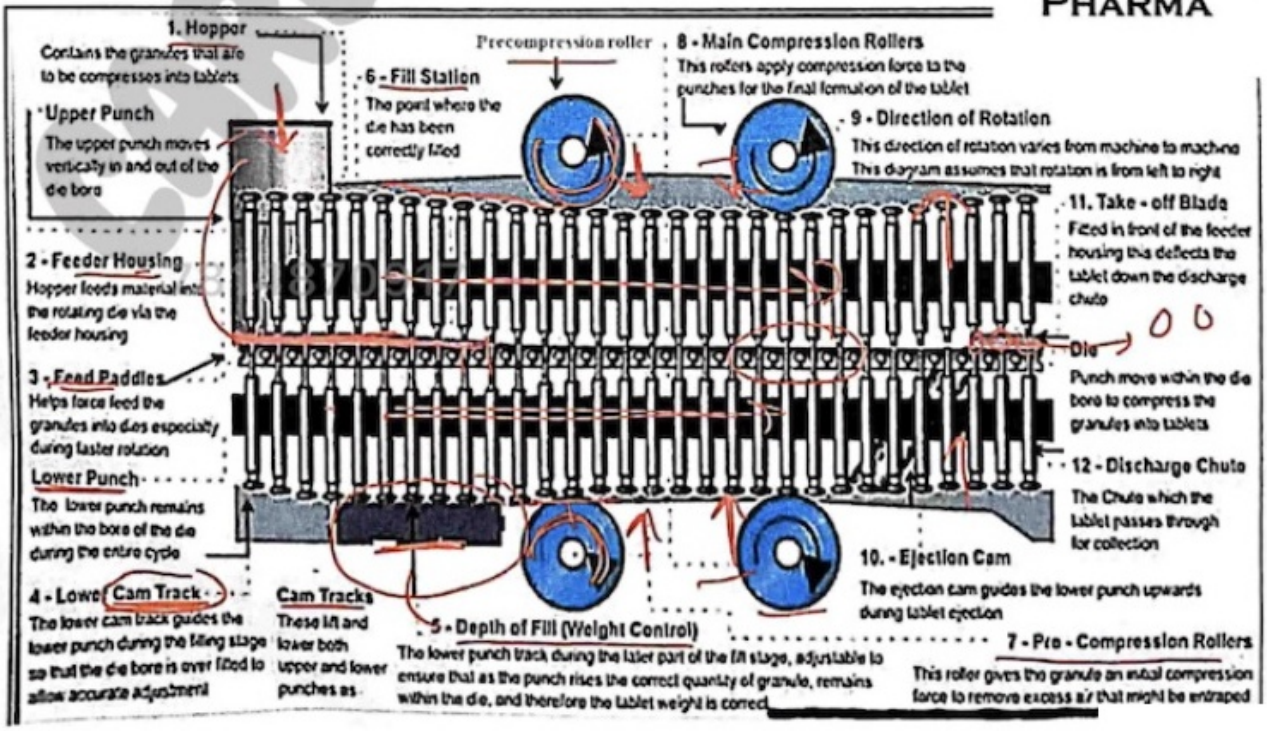


working
- feeding
- compression
- Ejection

TABLETS PROCESSING PROBLEMS

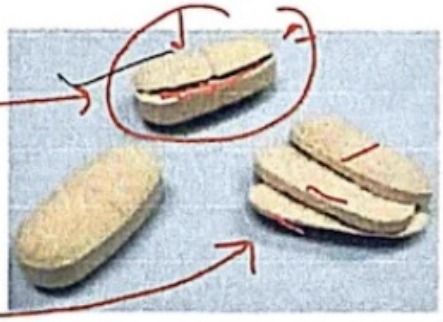
- Defects in tablets

Defects of Tablet:

a. **Capping:** the partially or completely separation of the top or bottom crown of the tablet from the main body of a tablet.

b. **Lamination:** The separation of a tablet into two or more distinct layers is called lamination.

Cause: During the compression process air is entrapped among the particle or granules and does not escape untill the compression pressure is release.



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- During compaction, particles undergo the sufficient plastic deformation to produce die wall pressure greater than can be relieved by elastic recovery when the punch pressure is removed
- Often deep concave punches produce tabets that cap.
- A granulation that is too dry tends to cap or laminate for lack of cohesion
- **Treatment**
- Settling the dies and punches properly
- Punches should be either buffed or polished before its use.
- Regulate the speed of tablet machine
- Regulate the pressure on the punches so as to maintain a proper degree of compression

2- Picking and sticking:

- The material is removed or picked up by upper punch from the upper surface of the tablet. In the sticking the material stick to the wall of the die cavity.

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- **Cause:** Use of worn out dies and punches
- Presence of moisture In the granules
- Excess of powder in the granules
- Scratches on the upper surface of the face of the punch
- **Treatment:**
- A new set of die and punches
- A proper quantity of lubrication in the granulation
- Dry granulation

3 Mottling:

- An unequal distribution of colour on the surface of a coloured tablet

- **Causes:**
- Drug whose colour differ from the tablet excipients or a drug whose degradation products are coloured
- A dye can cause mottling by migrating to the surface of a granulation during drying.
- The use of colourant in direct compression formulation can lead to mottling of the dye is not well dispersed or if its particle size is too large
- **Treatment**
- The use of colourant may solve the above problem.
- Change the solvent system, change binder system,
- Reduce drying temperature

4 Weight variation:

- Weight variation occur during the compression of granules in a tablet machine and the tablet do not have the uniform weight"

Cause

- **Granule size and size distribution before compression:** If the large granules are being used to fill a small die cavity, relatively few granules are required and difference of only a few granules around the average may represents a high percentage weight variation.
- **Poor flow:** The die fill process is based on a continuous uniform flow of granulation from the hopper through the feed frame. When the granules does not flow readily, it tends to move spasmodically through the feed frame so that some dies are incompletely filled. The use of glidant may improve the flow
- **Poor mixing:** Sometimes the lubricant and glidant are not thoroughly distributed. The flow of particle is then impaired and the granules do not move efficiently into the dies. However inadequate mixing at the stage can result in unstatisfactory granulation

Punch variation: When the lower punches are of unequal lengths, the differences may be minute or in inches, the fill in each die varies because the fill is volumetric. Only good punch and die controll program can provide tooling of uniform dimensions

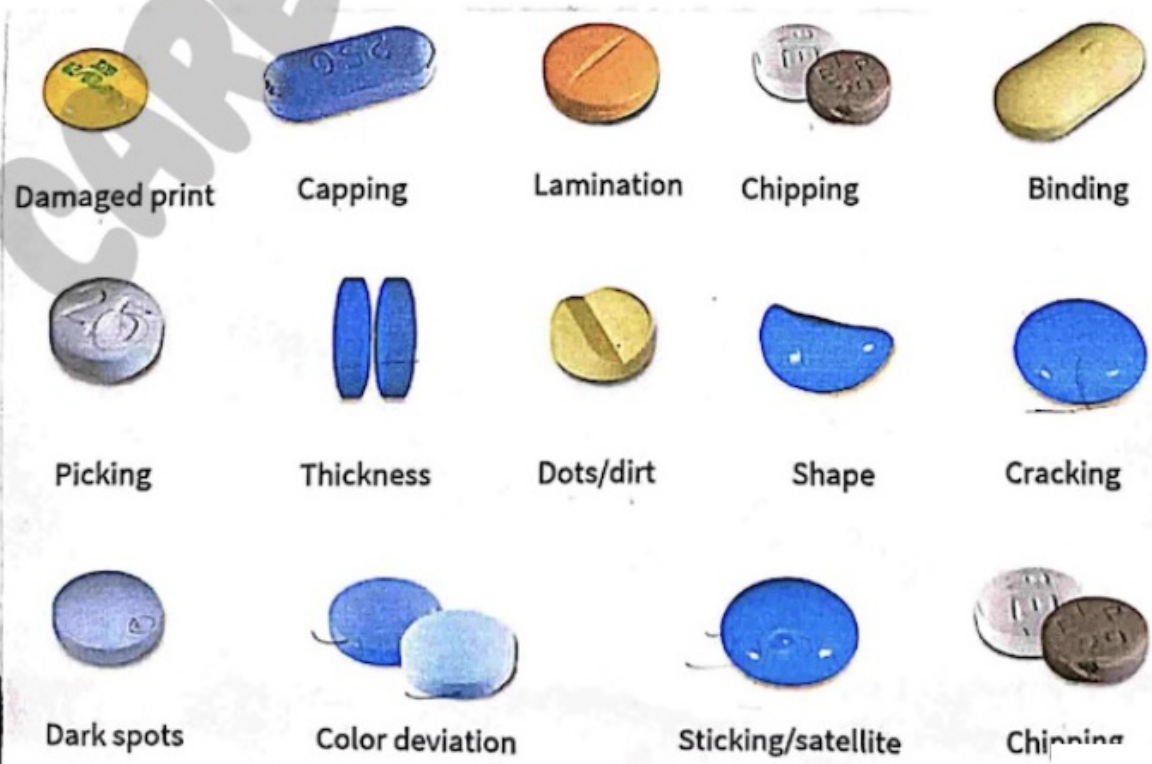
Hardness Variation

- The tablet do not have a uniform hardness. It depends on the weight of the material and space between the upper and lower punch during the stage of compression. If the volume of material or the distance between punches varies, hardness is likewise inconsistent.

5. **Double impression:**

- This effect occur when the lower punch has a monogram or some other engraving on it. During Compression, the tablet receives an imprint of the punch. Due to some defect in the machine lower punch move slightly upward before ejection of tablet and give second impression. This defect can be removed by controlling the undesirable movement of the lower punch.

6. - **Poor Flow:** Poor flow through the feed frame is usually a sign that the granulation is not flowing properly out of the hopper. As particulate solids move under the force of gravity through the progressively small openings they are subjected to uneven pressures from the mass above and alongside. Depending on the geometry of the hopper, this situation may give rise to one or another of two causes for poor flow i.e. "Arching" or "bridging" and "rat holing".
- **Treatment:** it may be controllable with vibrators attached to the hopper sides to induce the granulation flow.

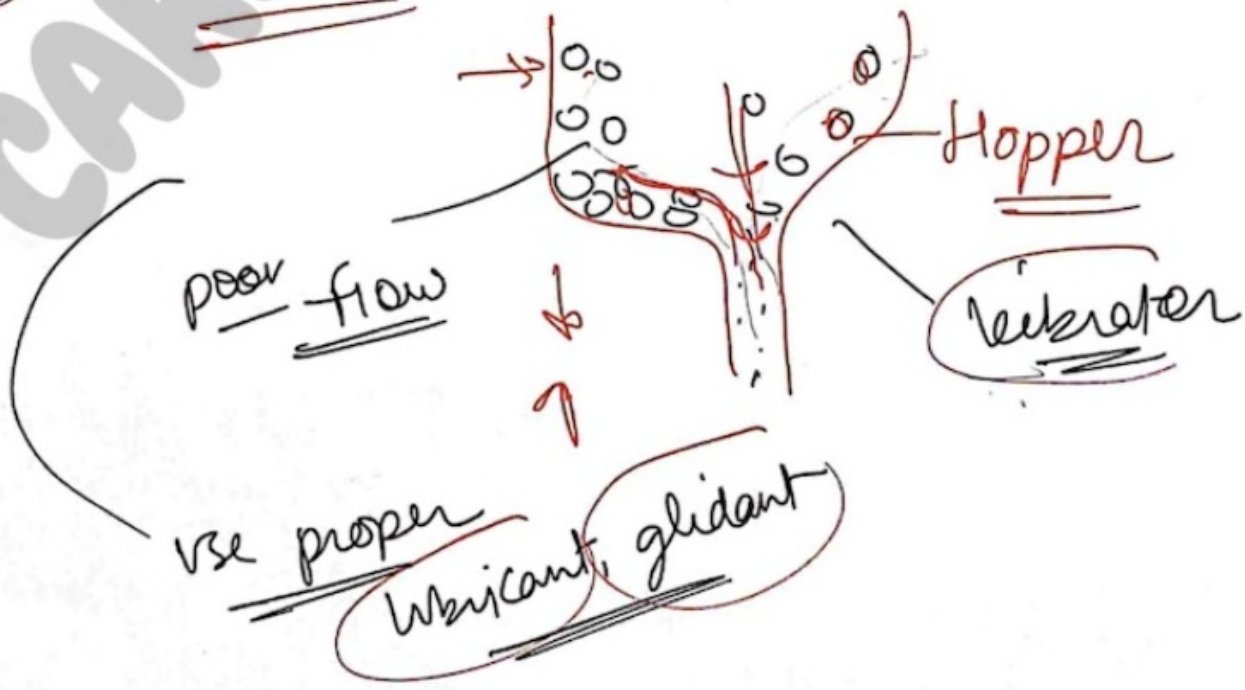


(5) double impression

double line

monogram

(6) Poor flow



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② Define tablet coating, write the types of coating. Note on-
- step involved in sugar coating with suitable eg.
- Film coating composition + Coating pan.

Tablet Coating - It is the process of applying a thin, uniform and continous layer of coating material on the surface of tablets.

- The coating may be sugar, polymer film or enteric material depending on the purpose.

- Coating is done to improve appearance, mask taste / odour, protect drug from environment and modify drug release.
- The process is usually performed in a coating pan or fluidized bed coater.

Causes / Reason of tablet Coating.

- Mask unpleasant taste, odor or color of the drug.
- Protect drug from environment factors.
- Increase mechanical strength
- Ease of swallowing.
- Improve appearance.

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Types of Coating - There are mainly three types of coating - They are as follows :-

(i) Sugar coating
(ii) Film coating
(iii) Enteric coating.

(i) **Sugar Coating** - These are those type of coating which is oldest and traditional method.

- It involve coating tablets with sugar syrup containing coloring agents and opacifiers. Makes tablet smooth, glossy and sweet. (make opaque (अपारदर्शी))
- The various steps involved in sugar coating ⇩
  (i) Sealing (water proofing)
  (ii) Subcoating
  (iii) Smoothing
  (iv) Coloring
  (v) Polishing.

Sugar Coated tablets

- In this coating, sugar layer is deposit on the surface of tablet
- It is used to mask the unpleasant taste of tablets & enhance acceptability of tablet.
  Eg. Brufen (Ibuprofen), Colofac (Mebeverine)

(i) **Sealing (water proofing)** - It is a seal coating over the tablets which prevents tablets from water penetration or moisture.

- Protect tablet core from moisture and make it hard.
- increase stability & safety of drugs.

Example - Shellac, Cellulose acetate phthalate (CAP) etc...
Zein ← A corn protein derivatives

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(ii) **Sub-coating** - This step of coating increase the weigh of the tablets.

- It increase size, round off edges, and mask core shape.
- Repeated several times until desired thickness.
- Also apply dusting powder to prevent sticking.

Calcium carbonate, Titanium dioxide.

Example - Gelatin, Acacia, Gum etc....

(iii) **Smoothing (Syruping)** - In this step smoothen rough subcoat surface and increase tablet bulb to desired size.

- Several coat of simple syrup solution (60-70%) are applied.
- May need 3-5 coats.

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Example - Concentrated sugar syrup without powder, pigments etc...

(iv) **Coloring** - It gives the uniform color and appearance. Sugar coating solution consist of 70% syrup and other coloring pigments.

- Add colorants / dyes to sugar syrup and apply in multiple thin layers.
- Water soluble dyes or water insoluble pigments may be used.

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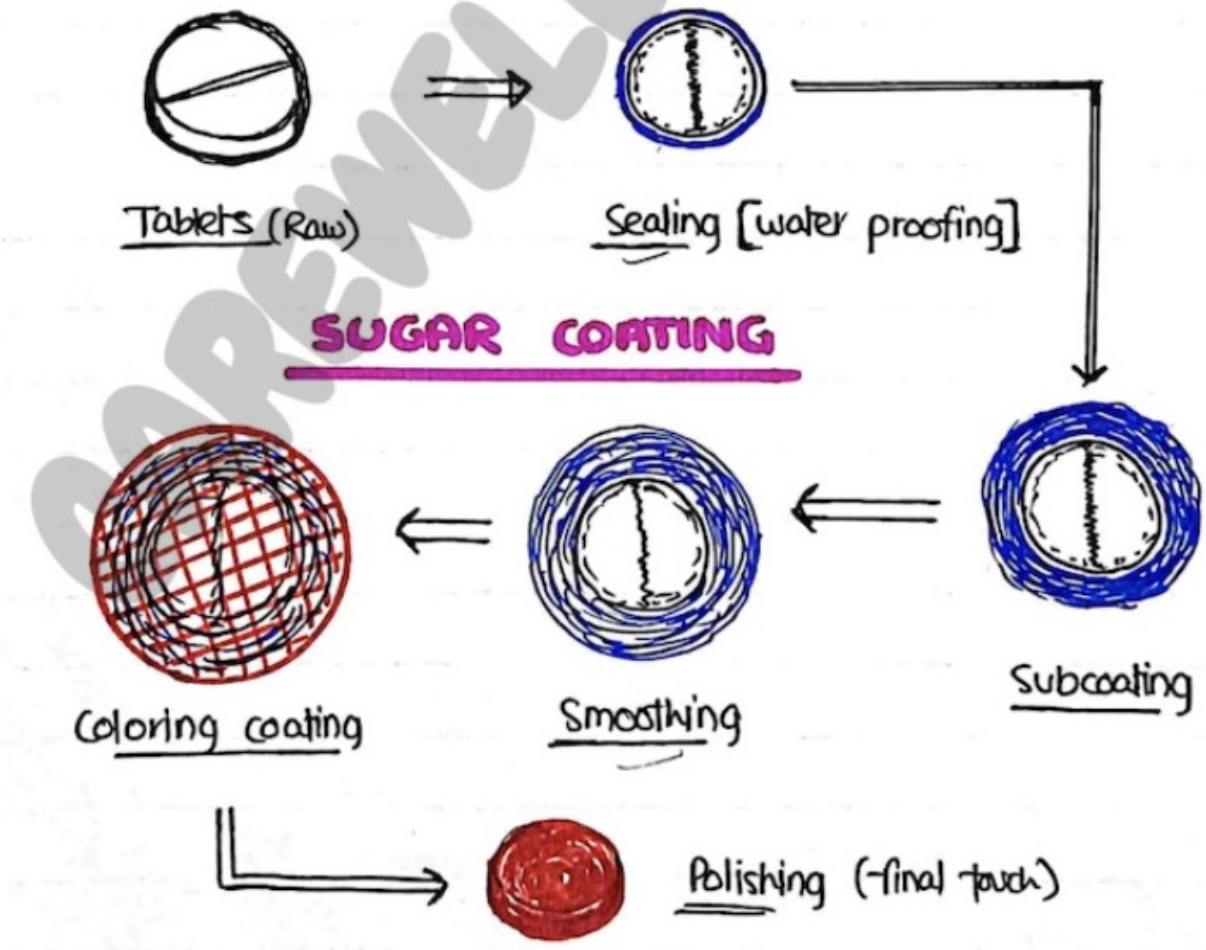
(v) **Polishing** - It provides Glossy and shiny finish. Generally sugar-coated tablets are dull in appearance. So polishing give the surface shine and tablet elegance.

Example - Beeswax, Candelila wax etc...

Sugar Coating ⟹ Tablet → (i) Sealing → (ii) Subcoating → (iii) Smoothing → (iv) Coloring → (v) Polishing



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② **Film coating** - It is a modern tablet coating technique in which a thin, uniform layer of polymer film is applied to the surface of a tablet

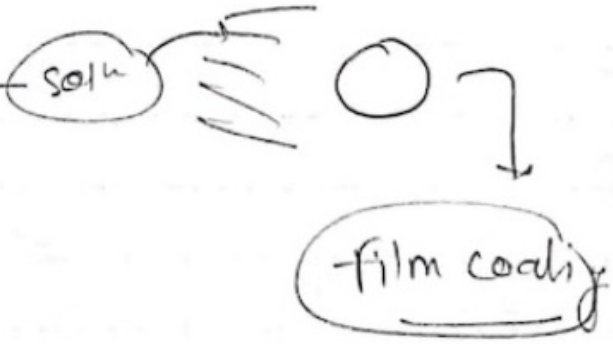
- It replaces the old sugar coating method because it is faster, lighter and less bulky.
- The thickness of the coat is usually 20 - 100 μm (very thin)

Advantages

- Faster and economical compared to sugar coating. (from atmospheric effects)
- Can be applied for taste masking, color, and protection.
- Better mechanical strength and less chance of coating defects.
- Improves swallowability and patient compliance.
- Produces thin, elegant and uniform coating.
- Suitable for moisture protection and light protection.

Material used in film coating ← various composition

- Polymer (film former)
- Solvents
- Plasticizers
- Colourants
- Miscellaneous agents.



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- **Polymer (film former)** - These are the main coating layer. It is used to improve product appearance and to protect the drug substance from atmospheric conditions.
  Example - Hydropropyl methyl cellulose (HPMC), Povidone etc...

- **Solvents** - It is used to dissolve or disperse polymer and carry it onto the tablet.
  Example - Water, Ethanol, Acetone etc...

- **Plasticizers** - It is used to improve flexibility and reduce brittleness of film.
  Example - Polyethylene glycol, Dibutyl phthalate.

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- **Colorants** - It is used to improve appearance and identification, protect from light.
  Example - Titanium dioxide, Iron oxides.

- **Miscellaneous agents** -
  - Surfactants ~ + flavors + sweeteners
  - Anti-adherents ~
  - Antioxidants etc. ~

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③ **Enteric Coating** - It is a special type of film coating in which the tablet is covered with polymer layers that resist disintegration in acidic pH (stomach) but dissolve in alkaline pH (intestine).

- To protect the drug from gastric juice OR to protect the stomach from irritation caused by the drug.
- The coating thickness is usually 50 - 200 µm.

Reasons -

- Protect acid - sensitive drugs from gastric acid (omeprazole)
- Release drug in intestine for local action.
- Prevent gastric irritation by drugs (eg. Aspirin).
- Improve bioavailability.
- to provide a delayed - release components for repeat - action tablets.

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Formulation of Coating composition - A typical enteric coating solution contains:

- Polymer - 5 - 10% (CAP or HPMCP)
- Plasticizer - 1 - 3% (PEG 400, dibutyl phthalate)
- Colorant / opacifier - 1 - 2 % (titanium dioxide).
- Solvents - q.s (water / ethanol / acetone).
- Additives - as required (anti - adherents, surfactants).

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- **Film coating** - In this type the thin film of polymer solution / dispersion sprayed on tablet surface
  - The thickness of this coating is usually about 20-100 µm
- **Dip coating** - This type of coating is applied by dipping core tablet in coating liquids and then dried.
- **Brush coating** - In this type, coating is applied by brushing (required experienced personnel).
- **Spraying** - Versatile, economical, fast coating process for different type, sized, shaped parts including large surface areas.
  - Coating is applied by spraying.
- **Electrostatic coating** - In this type, A substrate is present at which strong electrostatic charge is applied while coating material containing opposite charge is sprayed over it to get uniform coating.
- **Compression coating** - This is generally a tablet within a tablet, in which coating material is applied on the tablet core by compression using tablet machine
  - mainly used for mask the taste and to delayed the release of drug.

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Material used in Enteric Coating

Enteric coating polymer are (pH) sensitive polymers.

- insoluble at acidic pH (stomach)
- soluble at basic pH (small intestine).

Example - Cellulose acetate phthalate (CAP).

- Polyethylene glycol (PEG)
- Titanium dioxide
- Ethanol, acetone.
- Acylate polymers.

Methods of Coating

- Sugar coating
- Film coating
- Dip Coating
- Brush coating
- Spray coating
- Electrastatic coating
- Compression coating
- Vaccum film coating.

- **Sugar coating** - In this type of coating the deposition of sugar layer on the surface of tablet
  - It makes tablet smooth, glossy and sweet.

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- **Vaccum film coating** - In this type, tablet are placed in hot water jacketed pan, now vaccum is applied by sealing, then coating soln. is applied by hydraulic spray system.
- & heated pan remove maisture through evaporation.

Equipment employed in Coating.

There are three types of coating equipments which are as follows :-

(i) Standard coating pan
(ii) Perforated coating pan
(iii) fluidized bed coater (air suspension).

(i) **Standard Coating pan** - It is also known as conventional coating pan (pan coater)

- It is most popular because it is easy to operate and drug can be easily removed due to wide mouth of pan.

Construction & Working - It consist of a circular pan made of metal mounted on a stand.

- The pan is slightly tilted to an angle of about 45°C

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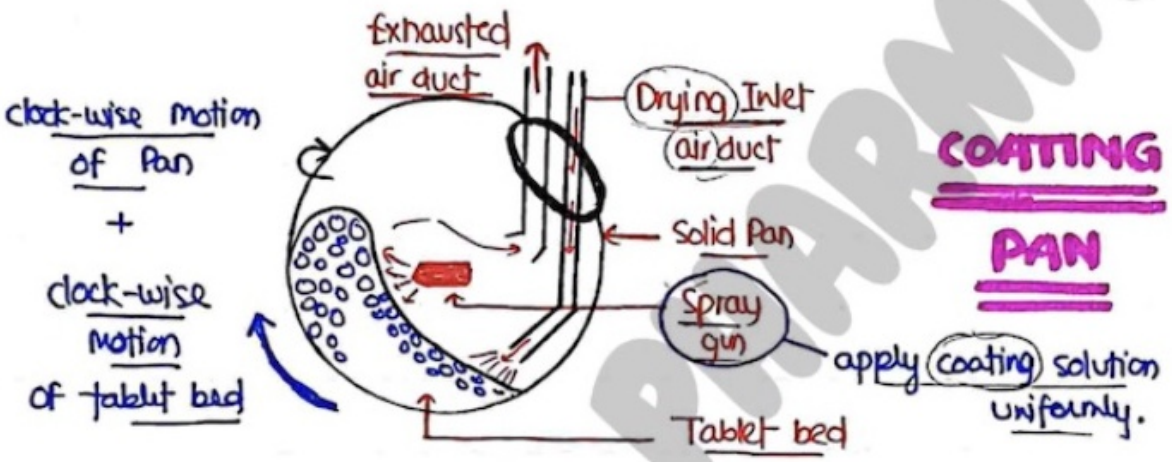
to the bench top. (↓ RPM ⇒ 10-20 rpm)

- The pan is about 30-80 inches in diameter.
- Now, tablets are introduced onto the tablet bed surface.



- Then, coating solution are applied to the tablets by spraying the material on to the rotating tablet bed.
- Also, The heated air is directed into the tablet bed by inlet air supply for drying of tablets.
- air leaves by exhausted duct.
- The temperature of hot air should be maintained.

Advantages →

- Simple to operate
- Suitable for both sugar and film coating
- Can handle large batch sizes

Uses →

- Sugar coating of tablets
- film coating for taste making, controlled release and protection.
- polishing of tablets.

Limitations →

- Conventional pans – slow drying, high time consumption
- Risk of uneven coating if not properly operated.

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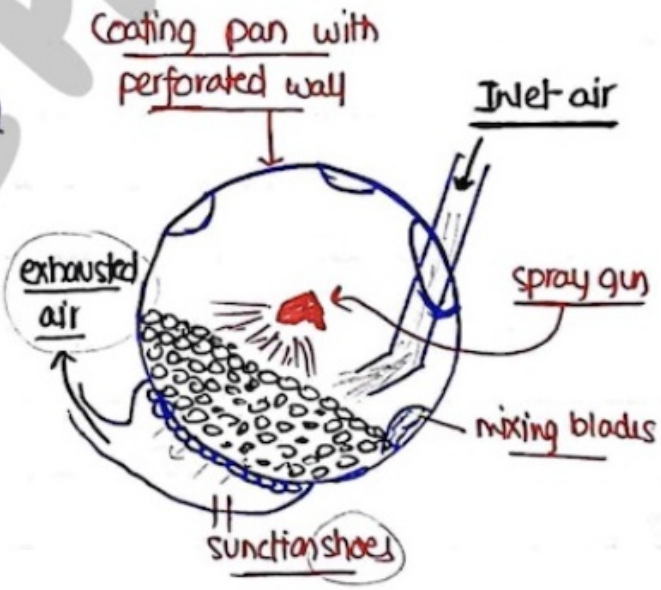
② **Perforated pan system** (↳ modern) - It consist of a perforated or partially perforated drum that is rotated on its horizontal axis in a (closed) chamber.

- The coating drum conduct various spray nozzles that atomize the coating soln.
- Drying air is passed through the tablet and is exhausted through perforations in the drum.
- Efficient drying system with high coating capacity [useful for both sugar & film coating].

Construction → It consists of :-

- Cylindrical / elliptical with perforated surface (stainless steel ↑ pan)
- Pan rotates horizontally with adjustable speed
- Spray gun
- Air handling system
- Exhaust system



Working / Principle → firstly, tablets loaded into the perforated pan, as pan rotates - tablet tumble & mix.

- Now, coating solution is sprayed by automatic spray gun
- Hot air enters through inlet, passes across the tablets
- Drying occurs rapidly & uniformly, then air exhausted through perforation.
- Process is repeated until desired coating is obtained.

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(Eg). Accela-cota , Hi-coater system , Glatt coater etc.

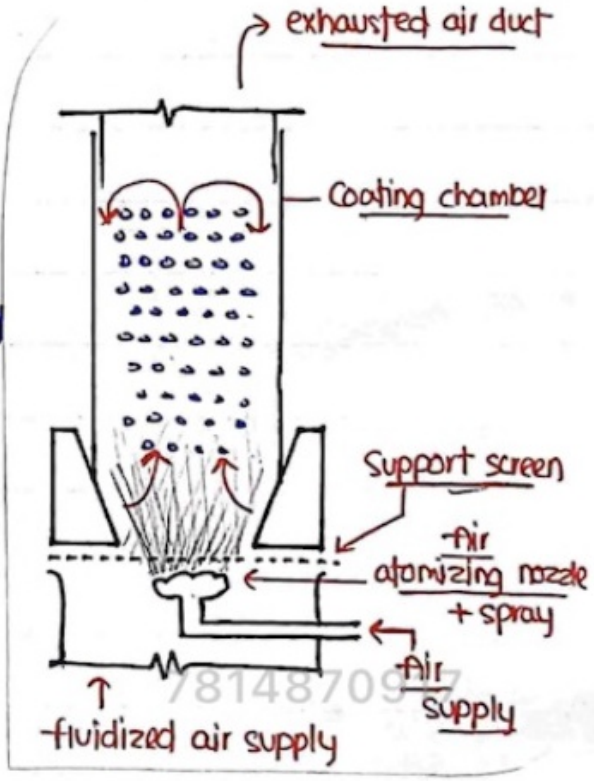
Advantages	Limitation	Uses
• Efficient drying • uniform coating • Reduced time • Automatic	• High initial cost • Requires skilled operators	• Sugar coating • film coating (most common) • functional coating (enteric, sustained release).

③ fluidised bed coater (Air suspension) → It is an advanced coating equipment.

- It work on the principle of fluidization (suspended in air).

Construction

- It consists of vertical cylinder.
- Air handling system
- Perforated container (perforated)
- Spray nozzle (spray coating soln)
  - top-spray, bottom-spray, tangential-spray
- Exhaust system (remove air/solvent)



Working

- firstly, tablets/pellets placed in product container
- High-velocity hot air is passed from below → particle become fluidized (suspended)
- coating solution is sprayed through nozzle → coats the particles.
- Hot air dries the wet surface immidiately, forms thin, uniform film.
- Process continues until desired coating thickness is achieved.

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- Now there are two types of spray system.
  (a) High pressure airless.
  (b) Low pressure air atomized.

(a) **High pressure airless** - Liquid is pumped at high pressure (250 - 3000 psig) through a small orifice [0.009 inch - 0.020 inch] in fluid nozzle which results in a finely divided spray.

(b) **Low pressure air atomized** - liquid is pumped through a some what larger orifice (0.020 inch - 0.60 inch) at relatively low pressure (5 - 50 psig).

- Low pressure air contact the liquid stream at the tip of the atomizer and finely divided spray is produced.

Advantages

- very uniform & smooth coating
- Rapid drying due to continuous hot air
- Can coat small particles, pellets, granules as well as tablets.
- Suitable for functional coating (enteric, sustained/controlled release).

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Limitation

- Expensive equipment
- Need high level of skill & process control

Uses

- film coating of tablets.
- Enteric coated
- Sustained/controlled release coating
- Taste masking of bitter drugs.

Defects in tablet Coating

In this we study about during coating, processing errors may lead to several defects in appearance or functionality of tablets.

- The problem occurs during and after coating the tablets are :-

(i) Blistering
(ii) Chipping
(iii) filling
(iv) Bridging
(v) Roughness
(vi) Blooming
(vii) Sticking and picking
(viii) Color variation.

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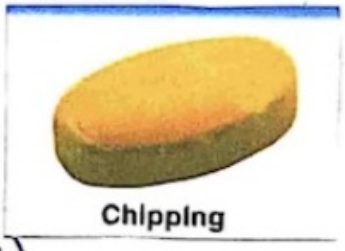
(i) **Blistering** - The high temperature occurs during drying process cause rapid evaporation of solvent and produce an un-smooth coated surface with number of uneven spots are called blisters.

Treatment -

- Lower drying temperature.
- Use coating solution with proper solvent system.

(ii) **Chipping** - It occurs when the film becomes dented and chipped from the edges of the tablet.

- The excess drying makes the coat brittle.
- Insufficient plasticizer in film.



Treatment -

- Add suitable plasticizer (Eg. glycerol, PEG).
- Avoid over-drying.

(iii) **Filling** - It is caused by applying too much solution, resulting in thick film that fills and narrows the bisect (fill monograph etc...).

Treatment - Careful monitoring of the fluid application rate and through mixing of the tablet in the pan prevent filling.

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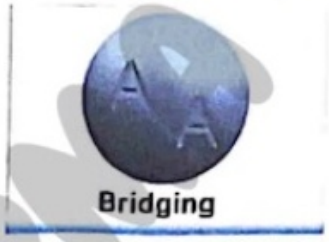
(iv) **Bridging** - During drying, the film may shrink and pull away from the sharp corners of an intagliation or bisect resulting in a bridging of the surface depression.

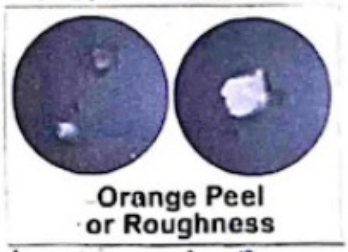
- Excess application of coating material.



Treatment -

- Adjust spray rate & viscosity of solution.
- Use appropriate plasticizer for flexibility.

(v) **Roughness** - A rough or gritty surface is observed when the coating is applied by spray. Some droplets may dry too rapidly before reaching the tablet bed cause deposition of spray dried particles.

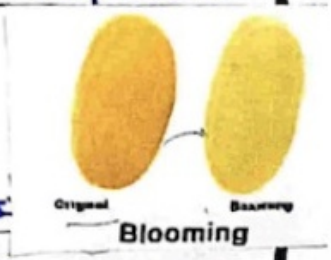


Treatment -

- moving the nozzle closer to the tablet bed or reducing the degree of atomization.

(vi) **Blooming** - The tablet become dull or hazed after prolonged priod of storage at high temperature.

- It is due to using too much plasticizer.



Treatment - Use correct type and concentration of ~~plas~~ plasticizer.

- Store at recommended temperature & humidity.

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(vii) **Sticking and Picking** - Over wetting and excessive film thickness causes tablets to each other or to the coating pan.

- Portion of film sticks to coating pan or other tablets. which leaves marks or bare areas.

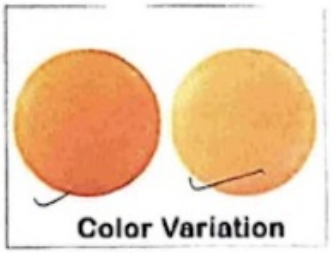
Sticking or Picking

Treatment -

- Reduce spray rate.
- Maintain proper drying conditions.

(viii) **Color Variation** - The difference in color intensity between tablets in a batch.

- It occurs due uneven mixing of coating suspension.
- Poor coating spray distribution.


Color Variation

Treatment -

- Stir coating solution continously.
- Ensures uniform spray pattern.

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③. Describe the various parameters of the in-process (IPQC) and finished products quality control tests for tablets.

evaluation of tablets

Quality Control Tests - These are the set of official (+ Non official) standards tests performed on a drug or dosage form (like tablets, capsules etc..)

- These tests are done to evaluate the each steps of formulation and also finished products to obtain a good quality of the products.
- There are several test for evalution of tablets :-

(I) Non-official tests :-

(i) General appearance – IPQC
(ii) Size & shape – IPQC
(iii) Thickness test – IPQC
(iv) Hardness test – IPQC
(v) friability test – IPQC

(II) Official test

(i) Weigh variation – IPQC
(ii) Content uniformity – FPQC
(iii) Dissolution test – FPQC
(iv) Disintegration test – FPQC

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(i) **General Appearance** - The tablet checked visually for color, shape, odour, surface finish, cracks, mottling, chipping, contamination.

- A good tablet must be uniform, elegant and free from defects.

← Elegant

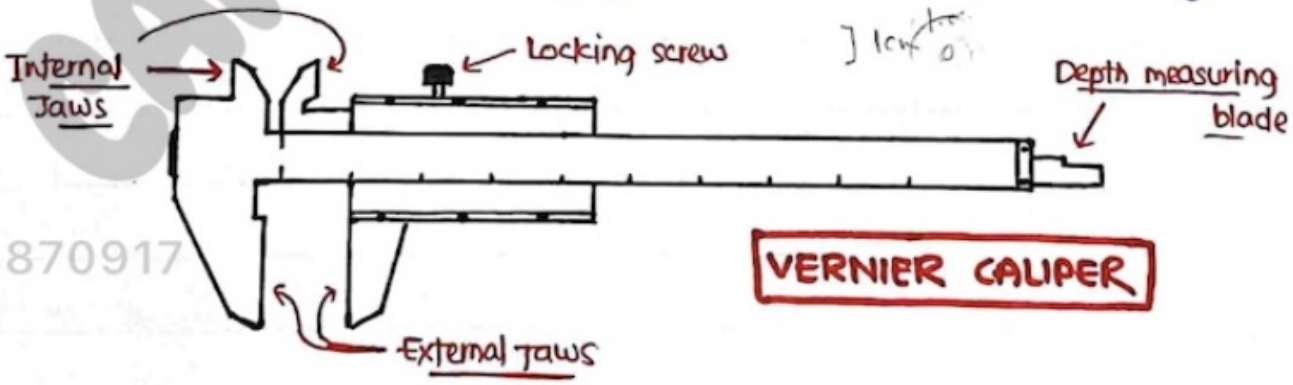
(ii) **Size & Shape** - It is measured by micrometer or vernier caliper.

- It is used maintains consistency in packaging, coating and dose accuracy.
- Generally they are biconvex, circular or flat.

← Circular

(iii) **Thickness** - It is checked using micrometer screw gauge. Its thickness variation should be ±5%.

- It ensures proper packaging and uniform film coating.



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(iv) **Hardness Test** - It measures tablet crushing strength (the force requires to break a tablet).

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- It is used to measures mechanical strength to resist handling during transport.
- It is measured by Monsanto, Pfizer or strong-Cobb tester.
- Ideal hardness – 4-8 kg/cm².

Monsanto

pfizer

- Too hard = poor disintegration
- Too soft = breakage during handling.

(v) **Friability test** - This test is official in BSP but not in I.P
Its main purpose is to check resistance to abrasion and breakage.

- This test is applicable for uncoated compressed tablets.
- To check physical strength of tablets during handling and transportation.
- The apparatus used in this test is Roche Friabilator.

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Procedure - weigh accurately 20 tablets randomly. (4min x 25 rpm)
- Placed in friabilator apparatus
- Rotated the apparatus at (25 rpm) for (100 revolutions)

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- Now Re- weighed the tablet and % of weight loss calculated.

Limit - Weight loss (<1%) = acceptable. ← variation (0.5-1%)

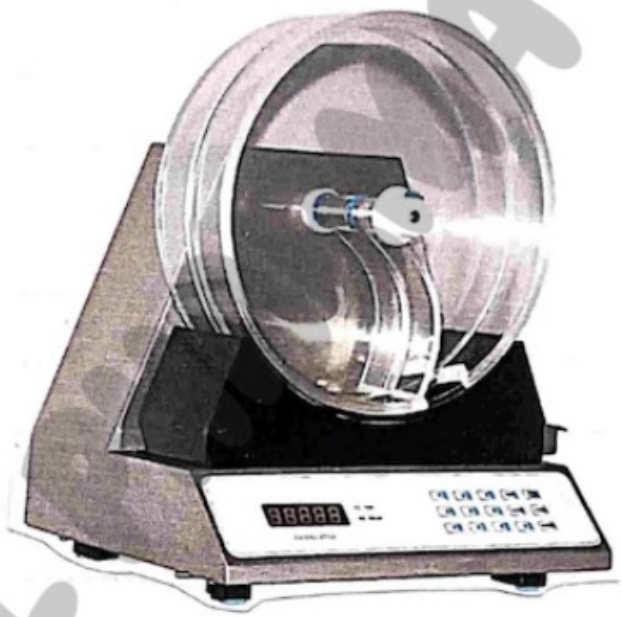
$$\% \text{ loss} = \frac{W_1 - W_2}{W_1} \times 100$$

where,

W_1 = initial weight

W_2 = final weight

- Variation is not more than 0.5-1%



(i) Weigh variation - Its main purpase is to ensures uniformity of drug dose.

- The weight of tablet measured routinely is help to ensure that tablet contain the proper amount of drug.

Procedure - Weigh individually (20) tablets.

- calculate the average weight.
- Each tablet weight compared with average.

As per (IP) limits -

Average weight of tablet	Permissible deviation.
(<80 mg)	(± 10%)
80-250 mg	(± 7.5%)
>250 mg	(± 5%)

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(ii) **Content uniformity** - Its main purpose is to ensures each tablet has the correct drug content.

- The 10 tablets assayed individually.
  ↳ drug content

Procedure - 10 tablets assayed individually

- Each should contain 85 - 115% of labeled claim with relative standard deviation ≤6%
- If these conditions are not met, then the tablet remaining from the 10 must be assay individually.

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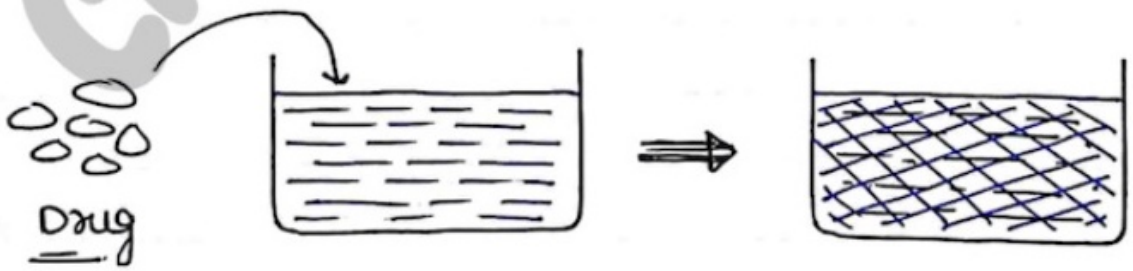
(iii) **Dissolution Test** - Dissolution is the process by which a solute dissolve into a solvent or solution.

- The main purpose is to measure rate and extent of drug release from tablet.
- It is also helps to determine the bioavailability of drug.



Apparatus - It consist of a cylindrical vessels with hemisphere bottom and made up of transparent glass having

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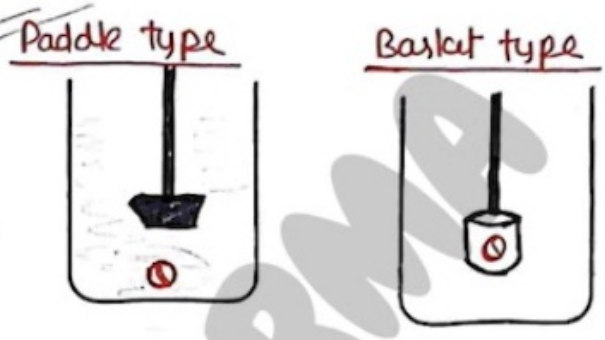
1000 ml capacity. The vessels is fitted with a cover having (4) holes, one for shaft, one for thermometer and rest two for sample.

- Acc. to I.P
  - Type I – Paddle type
  - Type II – Basket type.



DISSOLUTION TEST APPARATUS

Procedure - Place (1000 ml) of dissolution medium (dist. water), 0.1 N (HCl), Phosphate buffer) into vessel.

- Then this vessel immersed in water bath and maintain temp. (37°C) ± 0.5°C.
- Now placed tablet 6 according to type. [basket / paddle]
- Starts and speed maintain at 100 RPM or acc. to monograph (in I.P)

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- After specified time (5, 10, 15, 20, 45, 60 min.) interval (acc. to monograph in I.P) take 5-10 ml of sample (or specified) from vessels and refill vessels with 5-10 ml fresh medium.
- Now, this sample is tested by UV spectroscopy or some other suitable analytical method.
- Now, find the drug amount which released in sample and match with standards.

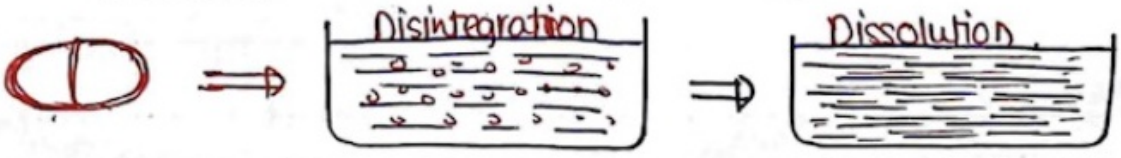
Interpretation → At the end of 30 min, 85% of the drug was released which meets the IP/USP specification. Hence the sample passes the dissolution test.

Sustained / Controlled release tablets → Drug release profile was found to be sustained upto 8 hours showing controlled release characteristics.

(iv)

Disintegration Test - It is the breaking up of a tablet into small fragments which is the first step before the drug dissolves and gets absorbed."

- The time required to disintegrate the tablet is called disintegration time.
- It is mainly used to check bioavailability & therapeutic efficacy.
- It is an official test under IP, BP and USP.



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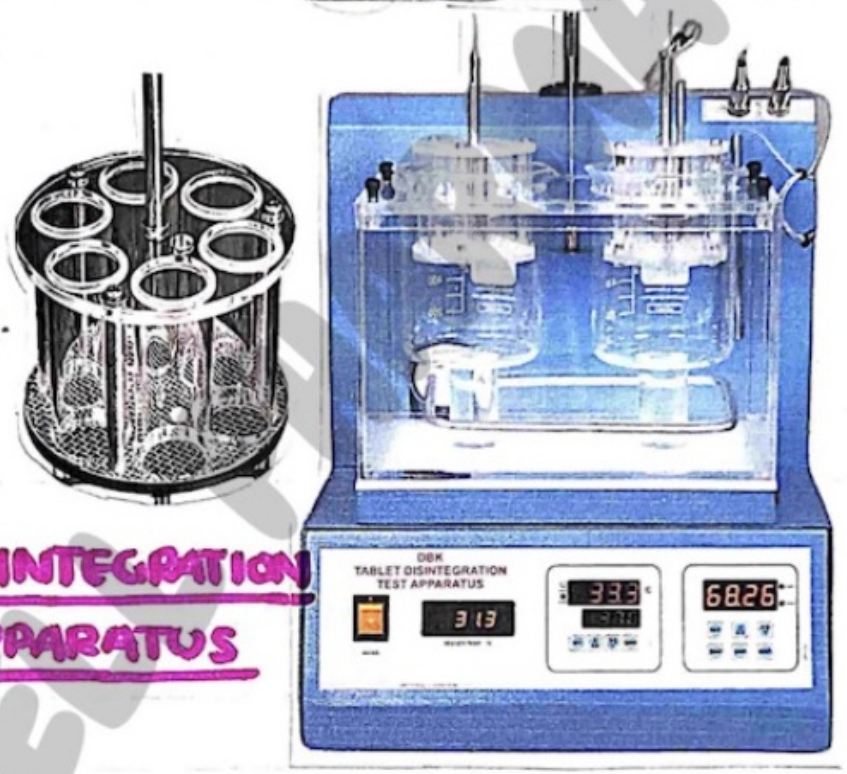
Apparatus - It consist of a water bath which is filled upto mark mentioned.

- A basket rack holding six plastic (or glass) tubes open at top and the bottom is covered with a 10 mesh (2 mm wire mesh) screen.
- Basket rack assembly moves up and down (28-32 cycles/min) in a beaker containing 800-1000 mL of medium.
- Temperature maintained at 37 ± 2°C (simulating body condition).



DISINTEGRATION APPARATUS

Procedure -

- The basket rack assembly is suspended in liquid medium in 1000 ml beaker and place this beaker into water bath and maintain the temperature of liquid at 37°C.
- Now, place one tablet into each tube.
- The assembly moves up and down at specified rate (30 RPM)
- A cylindrical disk made of transparent plastic is also placed over the tablet.
- The disc also generate little pressure on tablet.
- Now, noted down the disintegration time.
- Liquid Medium contains :→
  water (mostly), 0.1 N HCl (for enteric-coated tablets), Phosphate buffer pH 6.8

Uncoated Tablet	Disintegration Time
• Uncoated Tablet	within 15 minutes
• Sugar coated tablet	within 60 minutes
• Enteric coated tablet	within 60-120 minutes → in phosphate buffer pH 6.8
• Film coated tablet	within 30 minutes.

→ No disintegration in 0.1 N HCl for 2 hours

- Sublingual tablets → disintegrate within 2 min.
- Effervescent tablets → 5 min in water at room temp°.

Significance →

- Ensures batch-to-batch uniformity.
- Confirm that tablet will disintegrate properly in GIT for drug release.
- Failure indicates problem in formulation (binding excess, hard compression, coating defects, etc).

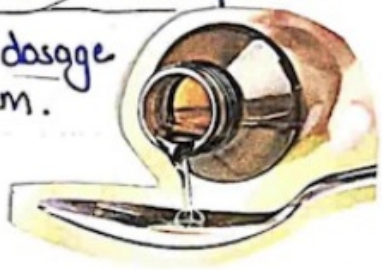
CAREWELL PHARMA

④. Define and classify liquid orals. It's ideal characterstics. Explain formulation, evaluation and manufacturing consideration of syrups, elixirs, suspension and emulsions.

Liquid dosage form — intended for oral route

Liquid Orals - These are pharmaceutical formulations intended for oral administration in a liquid state.

- They may contain one or more active pharmaceutical ingredients (APIs) dissolved, suspended or dispersed in a suitable vehicle.
- They are usually used for: older adults
  - → Pediatric and geriatric patients (who have difficulty children swallowing tablets or capsules).
  - → Faster onset of action due to quicker absorption.
  - → Flexible dosing compared to solid dosage form.



Ideal Properties -

- It should have an acceptable or pleasant taste.
- It should be Physically, chemically and microbiologically stable.
- It should be easy to swallow and dose.
- It should not irritate the gastrointestinal tract or oral mucosa.
- It should allow for adequate absorption of the drug into the bloodstream.
- Must meet microbial limits to ensure safety.
- It should be neither too thick nor too thin.

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Types of liquid orals - They are mainly of two types which are as follows :-

(i) Monophasic liquid orals → Syrup, Elixir.
(ii) Biphasic liquid orals → Suspension, Emulsion.

(i) **Monophasic** - These are single-phase, homogenous liquid preparations intended for oral administration.

- In this, the active pharmaceutical ingredients (API) is completely dissolved in a suitable solvent, usually water resulting in a clear solution.

Example - Syrup, Elixir.

(ii) **Biphasic** - These are two phase systems consisting of an insoluble drug dispersed or emulsified in a suitable liquid vehicle.

- These preparations are physically heterogeneous and the drug is not completely dissolved, but is present as solid particles or as immiscible liquid droplets.

Example - Suspension, Emulsions. → mixture of two immiscible liquid with the help of emulsifying agents.

Solid drug dispersed into liquid solvents.

Liquid drug dispersed in liquid solvent

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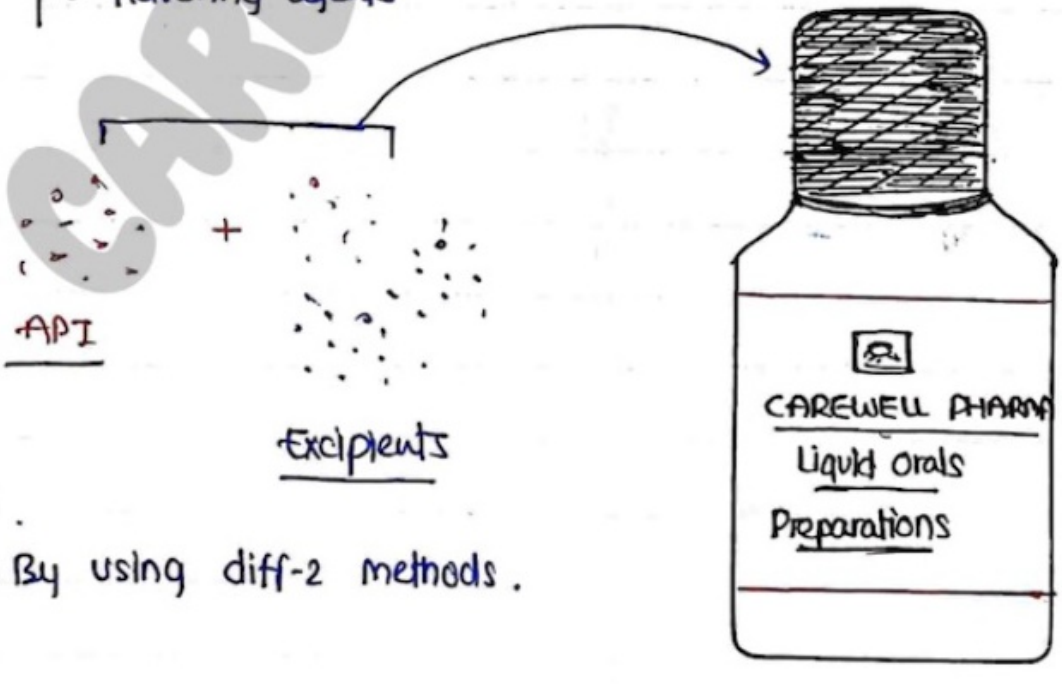
Formulation Consideration of liquid Oral Dosage forms.

- The goal of formulation is to develop a safe, stable palatable, effective and patient-compliant product. They must consider physiochemical properties of the drug, taste masking, solubility, stability and preservation.

- To formulate liquid orals, we required following excipients:—

  - API (If medicated)
  - Vehicles
  - Sweetening agents
  - Preservatives
  - Buffering agents
  - Antioxidants
  - Coloring agents
  - flavoring agents



- By using diff-2 methods.

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Vehicles - They serves as the base in which other components are dissolved / dispersed.

- Most commonly used Purified Water.

Sweetening Agents - Improve taste and patient acceptability, especially in pediatric formulations.

Example - Sucrase, Saccharin sodium.

Preservatives - Required to prevent microbial growth, especially in aqueous formulations.

Example - Methylparaben, Sodium benzoate.

Buffering Agents - Maintain the desired pH to ensure drug solubility, chemical stability, Preservatives efficacy

Example - Citric acid, Sodium citrate.

Antioxidants - Prevents oxidation of drugs and excipients.

Example - Ascorbic acid, Butylated hydroxyanisole (BHA).

Colouring Agents - Enhance aesthetic appeal and patient compliance. Non-toxic, approved by regulatory bodies.

Example - Tartrazine, sunset yellow.

Flavoring Agents - Used to mask unpleasant taste and odor of the drug should be compatible with the drug and stable formulation

Example - Fruit flavor, Mint.

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Syrup- A syrup is a sweet, viscous, concentrated aqueous solution that contains sugar (usually sucrose) with or without medicinal substances.

- They are commonly used as oral liquid dosage forms, especially for pediatric patients, due to their pleasant taste and ease of administration.
- The sucrose at the concentration of 85 % is dissolved in water.
  - 85 % w/v — acc to USP Simple syrup
- In syrup, sucrose contain 66.7 % w/w sugar.
  - 66.7 % w/w → Simple Syrup I.P.

Ideal Properties-

- Pleasant in taste and smell
- Attractive appearance
- Chemically and physically stable
- Microbiologically safe
- Non-toxic and non-irritant.

Types of Syrup

(i) Simple syrup
(ii) Medicated syrup
(iii) Flavoured syrup

(i) **Simple syrup**- It is a concentrated aqueous solution of sucrose (66.7 % w/w) in purified water, without any added medicinal agents.

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- Used as a sweetening agent in pharmaceutical formulations.

(ii) **Medicated Syrup** - It contains active pharmaceutical ingredients (APIs) along with sugar and other excipients.

- For therapeutic purposes, delivering drugs in liquid form, especially to children.
  Eg. Cough Syrup (Diphenhydramine).

(iii) **Flavoured Syrup** - It is a non-medicated sweetened, solution containing flavoring agents used mainly as a vehicle to mask unpleasant drug tastes.

- Helps in taste masking of bitter drugs.
  Eg. Orange syrup, Cherry syrup.

Formulation of Syrup

Ingredients	Purpose	Example.
• Active Drug (API)	Provides therapeutic effect	Diphenhydramine.
• Sugar (sucrose)	Sweetner, preservative	Sucrose (66.67% w/w)
• Purified water	Solvent or vehicle	Purified water B.P
• Coloring agents	Visual appeal (appearance)	Sunset yellow.
• Preservatives	Prevent microbial growth	Methylparaben.
• Viscosity enhances	Improve thickness and mouthfeel	Glycerin, Methylcellulose.
• Buffer	Maintain pH for stability	Sodium citrate.
• Flavors	Improve palatability	Orange, Raspberry.

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Method of Preparation

Agitation without heat

- Measure the required amount of purified water
- Add the sugar gradually while stirring continously
- Continue agitation until sugar is completely dissolved.
- Add heat-sensitive ingredients like (API), flavors, colorants etc...
- Make up the final volume with purified water.
- Filter if necessary and fill into containers.

Example - Vitamin syrup, Herbal syrup

Advantages - Suitable for thermolabile drugs.
- Prevents inversion of sucrose. which can cause sweetness loss and microbial growth.
- Preserves (volatile constituents (essential oils, aromatic flavors).

Disadvantages - (Slow) process.
- Requires constant and strong stirring.

Solution with heat

- Heat the required amount of purified water to around 80-85°C.
- Gradually add the sugar with constant stirring until dissolved.
- Cool the solution to room temperature.
- Add heat sensitive ingredients like (API), Preservatives, Colors.
- Adjust the final volume with purified water.
- Filter and transfer into suitable containers.

Example - Antacids, iron.

Advantages - Useful for (large scale)
- Quick and efficient dissolution
- Ensures a clear, concentrated syrup.

Disadvantages - flavors (lost) during heating.
- Not suitable for heat-sensitive drugs.

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Elixirs - These are clear, sweetened, hydroalcoholic oral liquid preparation intended for oral use, containing one or more active medicinal agents.

- They are designed to improve taste, stability and solubility of certain drugs that are not easily soluble in water alone.

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Ideal Properties

- To ensure patient acceptability.
- They should mask the unpleasant taste of drugs.
- They should be self-preserving due to alcohol content.
- They should not irritate the GIT.
- They should be Physically, chemically and microbiologically stable.
- They should be clear and free from turbidity or suspended particles.

Types of Elixirs -

(i) Medicated elixirs.

(ii) Non-Medicated elixirs.

(i) **Medicated elixirs** - These elixirs contain active pharmaceutical ingredients (APIs) and are used therapeutically to treat disease.

eg. Diphenhydramine, Antibiotic elixirs.

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(ii) **Non-Medicated Elixirs** - They do not contain any active drug. They are used as solvent or base in the preparation of medicated elixirs or as flavored vehicles to mask the taste of bitter drugs.

- It is used as diluent or flavoring base for extemporaneous preparations. acc. to prescriber

(eg). Alcohol, water.

Formulation

Component	Purpose
• Active ingredient (API)	→ Therapeutic agent
• Alcohol (ethanol)	→ solvent and preservatives.
• Water	→ Solvent
• Glycerin / Propylene glycol	→ Co-solvent and stabilizer.
• Sweetness	→ Taste masking
• Flavoring agents	→ Enhance palatability
• Coloring agents	→ Aesthetic appeal
• Preservatives	→ Prevent microbial growth.

Method of Preparation -

- Prepare aqueous solution (eg. sugars, water-soluble drugs).
- Prepare alcoholic solution (eg. alcohol-soluble drugs, flavors).

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- Now add the aqueous (water) solution to the alcoholic solution slowly with constantly (stir).
- If required, add glycerin or propylene glycol as co-solvents.
- Make up the final volume by adding (alcohol) or water (as required).
- Let the elixir stand for a few hours (usually (24) hours).
- Filter the solution to get a clear and sparkling elixir.
- Fill the elixir in airtight, light-resistant bottles.

Advantages:-

- Good in taste especially for children.
- Good solubility.
- Due to alcohol increasing the shelf-life and preventing microbial growth.
- Absorbed more quickly in the gastrointestinal tract.
- Helpful for pediatric or geriatric patients.

Disadvantages:-

- More expensive.
- Due to alcohol content not suitable for children, pregnant women.

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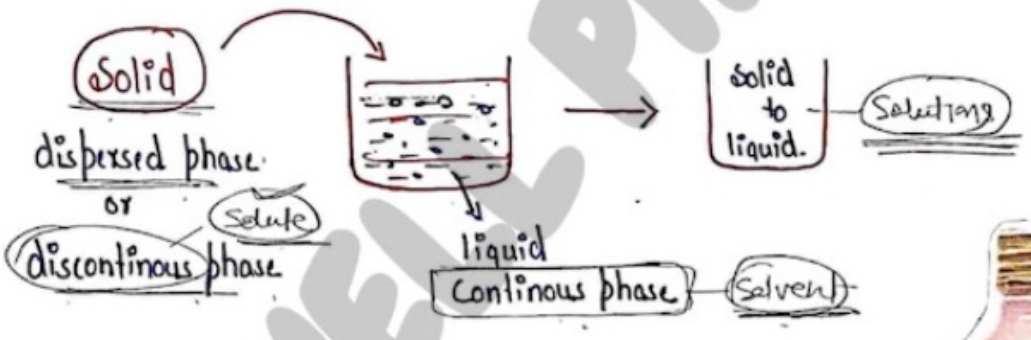
SUSPENSION

Suspension :- It is a (biaphasic) liquid dosage form of medicament in which the finely divided solid particles are suspended or (dispersed) in a liquid with the help of suspending agents.

- The solid particle act as (disperse) phase where as liquid vehicles act as the continous phase.
- It is ranging from 0.5 to 5.0 micron are dispersed in a liquid or semisolid vehicle.



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Advantages :-

- Higher rate of bioavailability.
- It is easy to swallow the insoluble medicament.

Disadvantages :-

- Need shaking before use.
- Accuracy of dosage is less.

Classification :- These are classified in (two) types :-

1. Flocculated Suspension
2. Deflocculated Suspension:-

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(i) Flocculated Suspension :- These agents produce binding between the particles to form aggregates or flocs, also have weak vander waal atraction forces.

(ii) Deflocculated Suspension :- These agents undergo aggregation at slower rate. They have shorter shelf life.

Excipients used in suspension and Method of Preparation.

(i) Wetting agents :- These are those agents which reduce the surface tension of water and increase the spreading abilities of a liquid.
(eg. Propylene Glycol USP.

(ii) Suspending agents :- These are the chemical additives that cause suspended solid to form aggregates called flocs which are easier to separates. It is widely used in water treatment plants.
(eg. Aluminum sulphate.

(iii) Thickener :- These are those substance which can increase the viscosity of a liquid without changing its other properties.
(eg. Polysaccharides.

(iv) Buffers and Preservatives :- These are those solution that can resist p^H change upon the addition of an acidic or basic components. They maintaing the (p^H) of the solution and stable.
(eg. Ammonium acetate.

Preservatives are those agents which prevents things from decaying / degradition / contamination, of microorganism.
(eg Nitrates.

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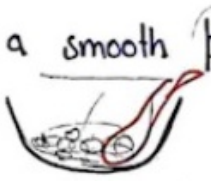
(v) Colouring agents :- These are those substance which are used to change the colour of a material (painting, printing)
(eg). Lead oxide.

(vi) flavouring agents :- These are those agents which are added to the food to mask the bitter taste.
(eg). Peppermint.,

Procedure :- [Method of Preparation]

- Grinding the insoluble materials in the mortor to a smooth paste with a vehicle containing the wetting agents.

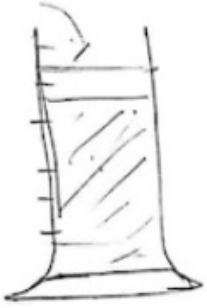


↓

- All soluble ingredients with same portion of vehicle are added to the smooth paste (step 1) to get slurry.

↓

- The slurry is transferred to a graduated cylinder, the mortor is rinsed with some part of vehicles.



↓

- Add suspending agent / flocculating agent.

↓

- finally adjust the volume by vehicle.

Stability Problems and Methods to overcome.

(1) Poor Re-Dispersibility :- The Dense mass of insoluble solid particles settle at the bottom which is known as caking.

Overcome :- Modification in the pH of the dispersion phase also prevent caking.

- Suitable (suspending agents) or mixture is added in the suspension.

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2. Cap Lacking :- It is a filling problem in which dispersed particl crystalline on the threads of the bottle cap and interfering during cap removal.

Overcome :- Different vehicles containing sucrose, glucose used in the preparation to overcome the problem.

3. Colour change :- Light sensitive preparation may change their colour in the presence of light due to increased surface area.

Overcome :- The bottle contaning light sensitive preparation should be kept in dark place.

4. Change in viscosity :- The long term storage of suspension alter their viscosity.

Overcome :- Add suitable viscosity agents in the Preparation.

5. Coking :- formation of hard cake. — flocs — soft c
- By adding flocculating agents. — suspending

6. p^H Change :- During long storage, p^H may changes.
- By adding buffering agents.

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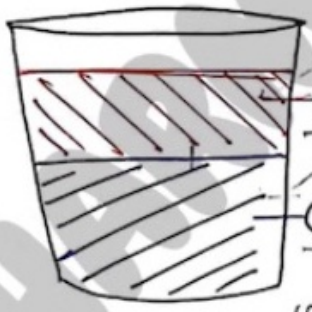
Emulsion :- An emulsion is a biphasic liquid dosage form in which two immiscible liquids are mixed together, to form a stable emulsion with help of emulsifying agent.

eg. Vanishing cream, Lotion, Gel.



- Advantages :-
  - More rapid absorbed.
  - Mask unpleasant taste
  - Easy to produce and scale up.
  - Sustained release medication.
  - High encapsulation efficiency

- Disadvantages :-
  - Chances of cracking emulsion
  - Dose accuracy decrease.
  - Unstable emulsion.

Solute + solvent = solution

(oil) (water) = O/W type

(water) (oil) = W/O type

dispersed phase

Types :-

(i) Oil in water :- These are those types of emulsion in which oil is a dispersed phase and water in continous phase. (O/W type)

eg. Milk, Creams.

(ii) Water in oil :- These are those type of emulsion in which water is mixed in oil or water is dispersed phase and oil is continous phase. (W/O type)

eg. Butter, Cold Cream.

- Emulsifying agents → These are those substance, which reduce the interfacial tension between two phase ie. aqueous & oil phase and produce stable emulsion. eg. surfactants
  - Naturally occuring agents (polysaccharides)
  - synthetic surfactants (benzalbonium chloride)

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- Method of Preparation of Emulsion.

(i) Dry gum method
(ii) Wet gum method
(iii) Bottle method.

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(i) Dry gum method :-

- In this method, the oil is first triturated with gum with a little amount of water to form the primary emulsion.
- The trituration is continued till a 'clicking' sound is neared and a thick white cream is formed.
- Once the primary emulsion is formed, the remaining quantity of water is slowly added to form the final emulsion.

∴ Ratio used [4:2:1] [oil : water : gum] → E.A

(ii) Wet gum method :-

- In this method, firstly gum and water are triturated together to form a mucilage.
- The required quantity of oil is then added gradually in small portions through trituration to form the primary emulsions.
- Remaining quantity of water is added to make the final emulsion.

∴ Ratio used [4:2:1] - [oil : water : Gum].

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3. Bottle Method :- [forbes method] :-

∴ used to prepare emulsion of volatile oils / or substance having very low viscosities.

1. part gum / acacia is placed in a dry bottle.
↓
2. part of oil are added.
↓
Shake the mixture throughly after capping.
↓
A volume of water (approx. equal to oil) is added in portions.
↓
Again shake the mixture throughly until the Primary emulsion is formed
↓
Dilute it with proper volume of water.

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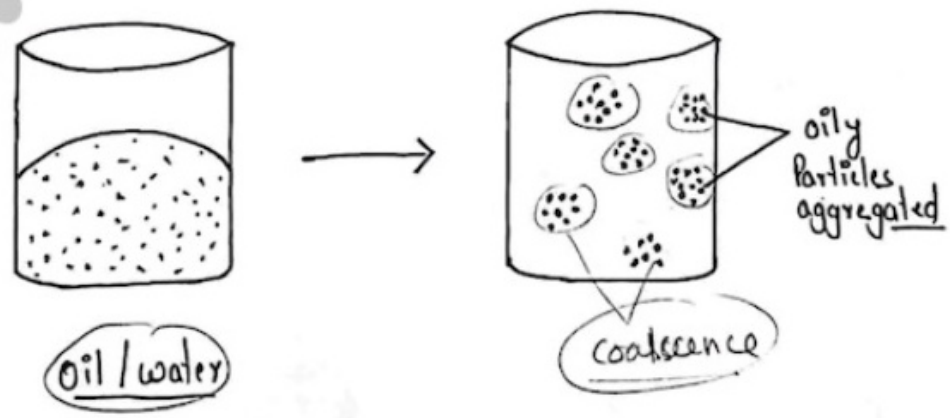
Stablity Problems and Methods to overcome :-

1. Coalescene :- The oily particles (Dispersed phase) because of sticky nature attract each other and they aggregate to form big particles.



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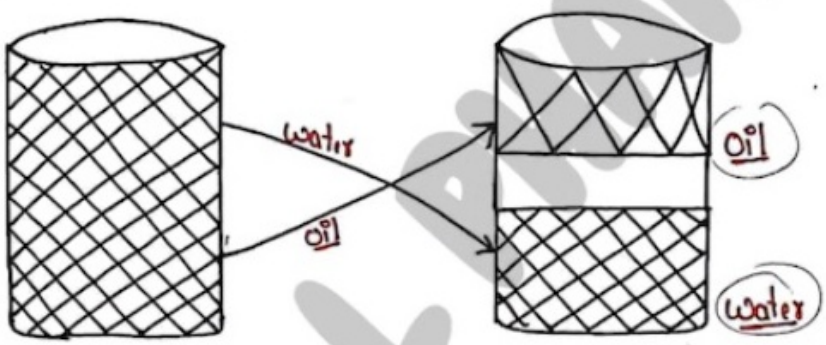
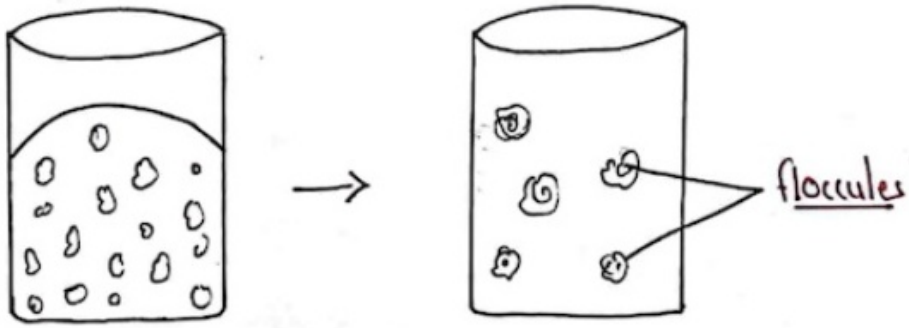
- **Overcome:-** Suitable emulsifying machinery should be employed in the preparation.
- Precaution should be undertaken while adding emulsifying agents.

② **Breaking (cracking):-** Due to improve mixing of oil and water in emulsion. Emulsion get separated into two layer oil and water.
- It is also occured due to improper ratio of oil and water.



- **Overcome:-** More emulsifying agents should be added in the preparation.
- The temperature at which the emulsion are store should be under control.
- Regulate the concentration of disperse phase. (o/w | w/o)

③ **flocculation:-** In this, due to increased surface free energy particles get aggregated to form floccules to decrease surface area.



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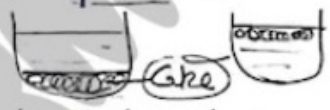
Overcome :- A high energy barrier exists in the presence of high charged density. On the dispersed droplets.

④ Creaming :- It is a reversible phenomenon. In which droplets of dispersed phase come together or deposite at the surface of the emulsion.



Overcome :- The viscosity of continous phase can be improved by adding viscosity enhancer.

- The density difference between the two phases should be reduced.
- The emulsion should be stored in a cool place at a low temperature

⑤ Phase inversions :- In this, the phase of emulsion get inverted

o/w → w/o / w/o → o/w

It happened due to mixing problem or by choosing wrong dispersed phase during formulation.

Overcome :- Add adequate quantity of preservatives in the preparation.

- Suitable preservative for each oil and aqueous phase should be used.

CARNIVAL PHARMA

Evaluation of Liquid Orals (official in Pharmacopoeia).

- Liquid orals include Syrup, solutions, Elixirs, suspension and Emulsions.
- They must be evaluated for quality, safety, stability and therapeutic efficacy.

(i) Organoleptic Properties -

- Appearance, colour, odour, taste → should be uniform and acceptable
- Free from turbidity, precipitation, phase separation or microbial growth.

(ii) Identification Tests -

- To confirm the presence of the active pharmaceutical ingredient (API).
- Methods - chemical reactions, UV/IR spectroscopy, TLC or HPLC fingerprinting.

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(iii) **Assay / Content of Active Ingredient** -

- Determines the exact drug strength in the preparation.
- Techniques - Titration, spectrophotometry, HPLC, depending on drug type.

(iv) **pH Measurement** -

- Ensures stability, solubility and palatability.
- Antitussives / Expectorants → slightly acidic.
- Measured using calibrated pH meter.

(v) **Uniformity of Content / Dose** -

- Particularly important for potent drugs in solution and suspensions.
- Each unit dose (5 ml) must contain nearly the same amount of drug.

(vi) **Viscosity Measurement** -

- Affects pouring, handling and dose uniformity.
- Measured using Brookfield viscometer or similar.

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(vii) **Microbial limit Test** -

- Must be free from pathogenic organisms like E. coli, Salmonella, Pseudomonas, S. aureus.
- Total viable count should be within specified pharmacopoeial limits.

(viii) **Stability Studies** -

- Physical stability - No change in colour, odour, taste, precipitation.
- Chemical stability - Drug content remains within limits.
- Microbiological stability - preservative effectiveness maintained.

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